

THE COLLECTED MATHEMATICAL PAPERS OF ARTHUR
CAYLEY
VOL. XII

SCAN PDF PAGES 176–225 (BOOK PP. 156–205; PAPER 825, CHS. IV CONCL.–VII)

Arthur Cayley (1821–1895)

Published 1889

*Transcribed from the original Cambridge University Press edition, 1889.
This file covers scan PDF pages 176–225 (book pages 156–205); restored from scan.*

Contents

825. A Memoir on the Abelian and Theta Functions (cont.)	2
Ch. IV concl. – Cubic Syzygy and Conversion	2
Chapter V. Miscellaneous Investigations	3
Chapter VI. The Nodal Quartic	32
Application of Abel’s Theorem	36
Chapter VII. The Addition Theory	39

825. A Memoir on the Abelian and Theta Functions (continued)

Chapter IV (concluded). The Fixed Curve a Cubic — Syzygy and Conversion.

The Fixed Curve a Cubic — Syzygy and Conversion. Art. Nos. 84 and 85.

[book p. 156] **84.** In the case when the fixed curve is a cubic (see Nos. 58 and 64), the analogous formulæ are

$$\frac{1}{2}\Omega(0; 1, 2; 3) = \frac{1 \cdot 2 \cdot 023 + 12 \cdot 031}{12s} = \frac{\{012\}}{12s} \{012\} \quad (\text{see No. 49}),$$

that is,

$$Q(0; 1, 2; 3) = \frac{\{012\}}{012} \cdot \frac{\{012\}}{12s} = \{012\} = \{12\overline{3}\},$$

where 1, 2 are the parametric points: 3 any other point on the cubic: the brackets $\{ \}$ are of course here necessary in order to distinguish $\{012\}$ from the determinant 012. It will be remembered that $\{012\}$ is an alternate function

$$\{012\} = -\{102\} = \{120\}, \text{ \&c.}$$

If instead of 1, 2 we take the parametric points to be α, β , coordinates $(x_\alpha, y_\alpha, z_\alpha)$ and $(x_\beta, y_\beta, z_\beta)$ respectively, then the formula is

$$Q_{\alpha\beta} = \frac{1}{2} Q(0; \alpha, \beta; 3) = \{0\alpha\beta\}\{3\alpha\beta\}.$$

Hence taking on the cubic a new point γ , coordinates $(x_\gamma, y_\gamma, z_\gamma)$ and forming the functions $Q_{\beta\gamma}$ and $Q_{\gamma\alpha}$ we have

$$Q_{\alpha\beta} + Q_{\beta\gamma} + Q_{\gamma\alpha} = \{0\beta\gamma\}\{3\beta\gamma\} + \{0\gamma\alpha\}\{3\gamma\alpha\} + \{0\alpha\beta\}\{3\alpha\beta\}.$$

But by the formula No. 58,

$$\{0\beta\gamma\} + \{0\gamma\alpha\} + \{0\alpha\beta\} = 0;$$

hence also

$$\{3\beta\gamma\} + \{3\gamma\alpha\} + \{3\alpha\beta\} = 0;$$

and we have thus

$$Q_{\alpha\beta} + Q_{\beta\gamma} + Q_{\gamma\alpha} = 0,$$

the syzygy for the cubic.

85. For the conversion, the definition of H is

$$H(0; 1, 2; 3, 6) = \partial_3 Q(0; 1, 2; 3) - \partial_6 Q(0; 3, 2; 6),$$

viz. this is

$$\begin{aligned} H &= H(0; 1, 2; 3, 6) = \partial_3(\{012\} - \{123\}) + \partial_6(\{132\} - \{326\}), \\ &= \partial_3\{012\} - (\partial_3 + \partial_6)\{123\} - \partial_6\{326\}, \end{aligned}$$

which, in virtue of $(\partial_1 + \partial_2 + \partial_3 + \partial_6)\{123\} = 0$ (see No. 63), becomes

$$H = H(1) = (\partial_1 + \partial_2)\{012\} - \partial_2\{123\} - \partial_6\{326\}.$$

[book p. 157] Interchanging the 0 and 1, we thence have

$$H_1(0) = \partial_0\{102\} + \partial_2\{023\} - \partial_6\{326\}.$$

Hence the difference $H_1(1) - H_1(0)$ is

$$= \partial_1\{012\} - \partial_0\{102\} + \partial_2\{123\} - \partial_2\{023\},$$

viz. this is

$$= (\partial_1 - \partial_0)\{012\} + \partial_2(\{123\} - \{023\}),$$

where the first term is

$$= -\partial_2\{012\},$$

and the whole therefore is

$$\begin{aligned} &= \partial_2(\{123\} - \{230\} - \{012\}), \\ &= -\partial_2\{301\}, \end{aligned}$$

in virtue of

$$\{123\} - \{230\} + \{301\} - \{012\} = 0;$$

the whole is consequently $= 0$.

We have thus

$$H_1(1) - H_1(0) = 0,$$

the conversion-equation in the case of the cubic.

Chapter V. Miscellaneous Investigations.

The Differential Symbol $d\omega$. Art. Nos. 86 and 87.

86. The definition is

$$\frac{y \, dz - z \, dy}{\frac{df}{dx}} = \frac{z \, dx - x \, dz}{\frac{df}{dy}} = \frac{x \, dy - y \, dx}{\frac{df}{dz}} = d\omega,$$

and it hence follows that we have

$$d\omega = \frac{\begin{vmatrix} dx & dy & dz \\ x & y & z \\ \lambda & \mu & \nu \end{vmatrix}}{\lambda \frac{df}{dx} + \mu \frac{df}{dy} + \nu \frac{df}{dz}},$$

where (λ, μ, ν) are arbitrary constants or, if we please, arbitrary functions of (x, y, z) : viz. the expression just written down is altogether independent of the values of λ, μ, ν , and is consequently equal to the value obtained by writing any two of these symbols $= 0$, that is, the expression is equal to any one of the foregoing three equal values of $d\omega$. The expression was first given by Aronhold (1863), in the memoir presently referred to.

[book p. 158] It is to be remarked that, considering (λ, μ, ν) as the coordinates of a point, the denominator $\lambda \frac{df}{dx} + \mu \frac{df}{dy} + \nu \frac{df}{dz}$ equated to 0 is the polar $(n-1)$ thic of the point λ, μ, ν in regard to the fixed curve.

If instead of λ, μ, ν we write $bc' - b'c, ca' - c'a, ab' - a'b$, where (a, b, c) (a', b', c') are constants, then the numerator is

$$= (ax + by + cz)(a' \, dx + b' \, dy + c' \, dz) - (a'x + b'y + c'z)(a \, dx + b \, dy + c \, dz),$$

or introducing ρ, σ to denote the arbitrary linear functions $ax + by + cz$ and $a'x + b'y + c'z$ respectively, the numerator is $= \rho d\sigma - \sigma d\rho$: moreover, observing that a, b, c and a', b', c' are the differential coefficients of ρ, σ in regard to the coordinates (x, y, z) , the denominator is $= J(f, \rho, \sigma)$; and the value of $d\omega$ is

$$d\omega = \frac{\rho d\sigma - \sigma d\rho}{J(f, \rho, \sigma)},$$

where, in accordance with a previous remark, the denominator equated to 0 is the polar $(n - 1)$ th of the intersection of the lines $\rho = 0, \sigma = 0$ in regard to the fixed curve.

Obviously, by taking for ρ, σ any two of the three coordinates x, y, z , we reproduce the original three forms of $d\omega$.

87. The last-mentioned form of $d\omega$ suggests the expression for this symbol in the case where the fixed curve, instead of being a plane curve, is a curve of double curvature defined by two equations $f = 0, g = 0$ between the four coordinates (x, y, z, w) : viz. ρ, σ being now arbitrary linear functions

$$ax + by + cz + dw, \quad \text{and} \quad a'x + b'y + c'z + d'w$$

of the four coordinates, the expression is

$$d\omega = \frac{\rho d\sigma - \sigma d\rho}{J(f, g, \rho, \sigma)};$$

and by taking for ρ, σ any two of the four coordinates x, y, z, w , we have for $d\omega$ six values which must of course be equal to each other; it is easy to verify

[book p. 159] *a posteriori* that this is so.

In the case where the curve of double curvature is not the complete intersection of two surfaces, the denominator (regarded as the Jacobian of the curve and of the arbitrary planes ρ, σ) will have a definite meaning, but what this is I do not at present consider.

The last-mentioned expression for $d\omega$ will be applied further on to the case of the quadri-quadric curve.

Integral Formulæ. Art. Nos. 88 to 90.

88. In what precedes, $d\omega$ has been used as a single symbol to denote any one of the equal differential expressions

$$\frac{y dz - z dy}{\frac{df}{dx}} = \frac{z dx - x dz}{\frac{df}{dy}} = \frac{x dy - y dx}{\frac{df}{dz}};$$

there is no quantity ω . The expressions are of the order $-(n - 3)$ in the coordinates (x, y, z) , and since (x, y, z) are as to their absolute magnitudes altogether arbitrary (only their ratios being determinate), a symbol such as

$$\omega = \int d\omega = \int \frac{y dz - z dy}{\frac{df}{dx}},$$

would, except in the case $n = 3$, be altogether meaningless. In fact, the integral would be

$$\int z^{n-3} \frac{d\left(\frac{y}{z}\right)}{\rho\left(\frac{y}{z}, \frac{x}{z}\right)} = \int \frac{d\left(\frac{y}{z}\right)}{z^{-(n-3)} \rho\left(\frac{y}{z}, \frac{x}{z}\right)},$$

where $\frac{x}{z}$ is, by the equation of the fixed curve, given as a function of $\frac{y}{z}$; but the other factor z^{n-3} is an absolutely indeterminate variable value, and the expression is meaningless.

But we have integrals $\int Q d\omega$, where Q is a homogeneous function of the order $n-3$ in the coordinates (x, y, z) ; and, in particular, we have such integrals where (corresponding to the forms which present themselves in the differential pure and affected theorems respectively) Q is either a rational and integral function $(x, y, z)^{n-3}$, or a rational and integral function $(x, y, z)^{n-2}$ divided by a linear function (x, y, z) : for in every such case, the form of integral is

$$\int \frac{d\left(\frac{y}{z}\right)}{\Phi\left(\frac{y}{z}, \frac{x}{z}\right)},$$

where $\frac{x}{z}$ is a given function of $\frac{y}{z}$, and the factor of $d\left(\frac{y}{z}\right)$ is thus a mere function of $\frac{y}{z}$. More definitely, in the integrals $\int Q d\omega$ which are considered, Q is either a minor function $(x, y, z)^{n-3}$, or it is the quotient of a major function $(x, y, z)^{n-2}$ by the linear function 012.

[book p. 160] In the case $n=2$, there is no rational and integral function $(x, y, z)^{n-3}$, but the function may be of the form belonging to the affected theorem, viz. it is unity divided by a linear function $(x, y, z)^2$; or say the integral is $\int \frac{d\omega}{ax + \beta y + \gamma z}$, where the (x, y, z) are connected by a quadric equation $(a, \dots)(x, y, z)^2 = 0$: it will be shown presently that this integral is obtainable as a logarithmic function.

In the case $n=3$, we have the rational and integral function $(x, y, z)^{n-3}$, a constant, or say $=1$, so that there is here an integral $\int d\omega$: we do not call this ω , but introducing a new letter, say u , and fixing at pleasure the inferior limit of the integral, we write $u = \int d\omega$.

89. In the foregoing form $\int Q d\omega$, so long as we retain the symbol $d\omega$, there is nothing to show what is the variable in regard to which the integration is to be performed; we may, for instance, writing

$$d\omega = \frac{y dz - z dy}{\frac{df}{dx}},$$

make it to be z , or in like manner to be any other of the six quotients. We thus cannot attribute a value to the inferior or superior limit of such an integral, but we may take the limits to be each of them a point on the fixed curve: for instance, if 0, 1 be points on the fixed curve, then the integral $\int_0^1 Q d\omega$ means the integral taken from the value at the point 0 to the value at the point 1 of the variable in regard to which the integration is performed; or when there is no expressed superior limit, then the integral is to be taken from the value for the expressed or known inferior limit to the value at the current point (x, y, z) of the variable in regard to which the integration is performed. The actual value of the integral will of course depend upon the path of the variable; but this is a question which is not here entered upon.

If using Cartesian Coordinates x, y , we write for instance

$$d\omega = \frac{dx}{\frac{df}{dy}}, \quad \text{then} \quad \int \frac{Q dx}{\frac{df}{dy}}$$

will denote an integral $\int \Phi x dx$ in regard to the variable x , and the inferior and superior limits will be as usual values of x ; or if there is no expressed superior limit, then the integral will be $\int_0^x \Phi x dx$ the integral taken from the inferior limit x_0 to the current value x .

[book p. 161] We may, if we please, consider the coordinates (x, y, z) as depending upon a parameter θ , viz. the ratios $x : y : z$ may be regarded as given functions of θ , and the integral $\int Q d\omega$ is then an integral $\int \Theta d\theta$, which taken from a constant inferior limit up to the value θ_1 belonging to a given point 1 of the curve, is a given function of θ_1 , or say of the point 1. But except in the case of the cubic (or generally $p = 1$), we do not have the coordinates actually given as known functions of a parameter θ (say they are potentially known functions of θ), and it is further to be noticed that the functions which present themselves are functions not of a single point, but of p or more points: thus in the case of the quartic, $n = 4$, $p = 3$; we have $\int_0^1 x d\omega$, $\int_0^1 y d\omega$, $\int_0^1 z d\omega$, each standing for a given function of the parameter θ_1 , but these integrals do not present themselves singly, but in combinations such as

$$\int_0^1 + \int_0^2 + \int_0^3 (x d\omega, y d\omega, z d\omega),$$

say these sums of integrals are u, v, w : each of the functions u, v, w is a potentially known function of the parameters $\theta_1, \theta_2, \theta_3$, which belong to the points 1, 2, 3, respectively, and is consequently regarded as a given function of these four points.

90. Consider as before, in the case of a cubic curve, the integral $u = \int d\omega$: it will presently be seen that for the general curve as given by a cubic equation $f = 0$ of any form whatever, we arrive at a form of elliptic function: but the ordinary elliptic functions $\text{sn}, \text{cn}, \text{dn}$ connect themselves most readily with the cubic curve $y^2 = x \cdot 1 - x \cdot 1 - k^2 x$. We have here

$$d\omega = \frac{dx}{2y} = \frac{dx}{2\sqrt{x \cdot 1 - x \cdot 1 - k^2 x}},$$

or, in the equation $u = \int d\omega$, taking the inferior limit to be 0, say

$$u = \int_0^x \frac{dx}{2\sqrt{x \cdot 1 - x \cdot 1 - k^2 x}},$$

an equation which determines u as a function of x , or conversely, x as a function of u . We might thence, by means of Abel's theorem as applied to the curve in question, investigate the properties of the function $x = \lambda(u)$ thus arising, and so establish the theory of elliptic functions: but it is more convenient, treating the elliptic functions as known functions, to write for λu its value; viz. to take for x as given by this equation, the value $x = \text{sn}^2 u$: we thence have $y = \text{sn} u \text{ cn} u \text{ dn} u$; viz. these values $x = \text{sn}^2 u$, $y = \text{sn} u \text{ cn} u \text{ dn} u$, satisfy the equation $y^2 = x \cdot 1 - x \cdot 1 - k^2 x$ of the curve, and give, moreover, $d\omega = du = \frac{dx}{2y}$: and we can with these values, and the formulæ for elliptic functions, verify any results given by Abel's theorem. This will be done in considerable detail: but at present I wish only to remark that the formulæ give the coordinates x, y of a point on the cubic curve expressed as one-valued functions of

C. XII.

21

[book p. 162] a parameter or argument u : but that this argument u is not a one-valued function of the coordinate x , or even of the coordinates x, y of the given point on the curve: say the argument u has not a unique value for a given point (x, y) of the curve. There are, in fact, an infinity of values $u = u + 2mK + 2m'iK'$, where m, m' are any positive or negative integers: that this is so, depends on the multiplicity of values, according to the different paths of the variable, of the integral

$$\int \frac{dx}{2\sqrt{x \cdot 1 - x \cdot 1 - k^2 x}},$$

or, regarding the elliptic functions as known functions, it depends upon the double periodicity of these functions.

Aronhold's Quadric Integral. Art. Nos. 91 to 93.

91. I reproduce the investigation contained in Aronhold's paper "*Ueber eine neue algebraische Behandlungsweise u.s.w.*," Crelle, t. LXI. (1863), pp. 95–145. We take the general quadric function $(a, b, c, f, g, h)(x, y, z)^2$: $\alpha x + \beta y + \gamma z$ an arbitrary linear function of x, y, z : the theorem is $\frac{d\omega}{\alpha x + \beta y + \gamma z} =$ differential of logarithm of an algebraic function of (x, y, z) ; viz. taking (ξ, η, ζ) for the coordinates of either of the points of intersection of the line $\alpha x + \beta y + \gamma z = 0$ with the quadric $(a, \dots)(x, y, z)^2 = 0$, and writing also

$$H = 2(bc - f^2, ca - g^2, ab - h^2, gh - af, hf - bg, fg - ch)(\alpha, \beta, \gamma)^2,$$

then the theorem is

$$\frac{d\omega}{\alpha x + \beta y + \gamma z} = \frac{1}{\sqrt{H}} d \log \frac{(a, \dots)(x, y, z)(\xi, \eta, \zeta)}{\alpha x + \beta y + \gamma z},$$

or, what is the same thing,

$$\int \frac{d\omega}{\alpha x + \beta y + \gamma z} = \frac{1}{\sqrt{H}} \log \frac{(a, \dots)(x, y, z)(\xi, \eta, \zeta)}{\alpha x + \beta y + \gamma z} + \text{const.}$$

It is to be observed, in reference to this equation, that the two sides respectively are in regard to (α, β, γ) homogeneous functions of the degree -1 , and in regard to (ξ, η, ζ) homogeneous of the degree 0 ; viz. on the right-hand side the effect of a change in the absolute magnitudes of ξ, η, ζ , say the change into $k\xi, k\eta, k\zeta$, is merely to change by $\log k$ the constant of integration.

It is to be remarked also that the equation $(a, \dots)(\xi, \eta, \zeta)(x, y, z) = 0$ represents the tangent to the conic at the point (ξ, η, ζ) of intersection with the line $\alpha x + \beta y + \gamma z = 0$; calling the linear function in question T , the value of the integral is $\frac{1}{\sqrt{H}} \log \frac{T}{\alpha x + \beta y + \gamma z}$; the two (ξ_1, η_1, ζ_1) (ξ_2, η_2, ζ_2) are the coordinates of the points of intersection respectively, and the two values thus are $\frac{1}{\sqrt{H}} \log \frac{T_1}{\alpha x + \beta y + \gamma z}$, $\frac{1}{\sqrt{H}} \log \frac{T_2}{\alpha x + \beta y + \gamma z}$, then in passing from one of these to the other we change the sign of the radical $\sqrt{H}$, $\frac{1}{\sqrt{H}} \log \frac{T_1}{\alpha x + \beta y + \gamma z}$ and $-\frac{1}{\sqrt{H}} \log \frac{T_2}{\alpha x + \beta y + \gamma z}$. These must

[book p. 163] differ by a constant only: viz. we should have $\log \frac{T_1 T_2}{(\alpha x + \beta y + \gamma z)^2} = \text{a const.}$ And, in fact, T_1 and T_2 being the tangents to the conic f at its intersections with the line $\alpha x + \beta y + \gamma z = 0$, we have it is clear $f = \lambda T_1 T_2 + \mu(\alpha x + \beta y + \gamma z)^2$, that is, (x, y, z) referring to a point of the conic $f = 0$, we have $\frac{T_1 T_2}{(\alpha x + \beta y + \gamma z)^2} = \text{a constant}$, which is right.

92. We require the coordinates (ξ, η, ζ) of an intersection: these are determined by the equations $\alpha\xi + \beta\eta + \gamma\zeta = 0$, $(a, \dots)(\xi, \eta, \zeta)^2 = 0$, or as these may be written

$$\alpha\xi + \beta\eta + \gamma\zeta = 0,$$

$$\xi(a\xi + h\eta + g\zeta) + \eta(h\xi + b\eta + f\zeta) + \zeta(g\xi + f\eta + c\zeta) = 0;$$

we have thence ξ, η, ζ proportional to the determinants

$$\begin{array}{ccc} \alpha\xi + h\eta + g\zeta, & h\xi + b\eta + f\zeta, & g\xi + f\eta + c\zeta \\ \alpha, & \beta, & \gamma \end{array}$$

say these determinants are $\Omega\xi$, $\Omega\eta$, $\Omega\zeta$, where Ω is a value as yet undetermined. The equations are $\gamma(h\xi + b\eta + f\zeta) - \beta(g\xi + f\eta + c\zeta) = \Omega\xi$, &c., viz. these are

$$(\gamma h - \beta g)\xi + (\gamma b - \beta f - \Omega)\eta + (\gamma f - \beta c)\zeta = 0$$

$$(\alpha g - \gamma a - \Omega)\xi + (\alpha f - \gamma h)\eta + (\alpha c - \gamma g)\zeta = 0$$

$$(\beta a - \alpha h)\xi + (\beta h - \alpha b)\eta + (\beta g - \alpha f - \Omega)\zeta = 0;$$

eliminating (ξ, η, ζ) , we have an equation which may be written

$$\begin{vmatrix} A - \Omega, & B, & C \\ A', & B' - \Omega, & C' \\ A'', & B'', & C'' - \Omega \end{vmatrix} = 0,$$

that is,

$$\begin{vmatrix} A, & B, & C \\ A', & B', & C' \\ A'', & B'', & C'' \end{vmatrix} - \Omega(BC' - B'C + C'A'' - C''A' + A'B'' - A''B') + \Omega^2(A + B' + C'') - \Omega^3 = 0.$$

We find very easily that the determinant and $A + B' + C''$ are each $= 0$; and the equation thus reduces itself to

$$\Omega^2 = BC' - B'C + C'A'' - C''A' + A'B'' - A''B',$$

[book p. 164] **93.** We may now verify the theorem; in the general expression for $d\omega$ by λ, μ, ν the values ξ, η, ζ , the equation to be verified becomes

$$\frac{d\omega}{\alpha x + \beta y + \gamma z} = \frac{1}{\Omega} \frac{(\alpha, \dots)(\xi, \eta, \zeta)(dx, dy, dz)}{(\alpha x + \beta y + \gamma z)(\alpha, \dots)(\xi, \eta, \zeta)(x, y, z)},$$

and, in fact, each of the three lines is separately $= 0$, so the equation to be verified becomes

$$\frac{1}{(ax + hy + gz, \dots)(\xi, \eta, \zeta)}(dx, dy, dz) = \frac{1}{\Omega} \frac{N}{(\alpha x + \beta y + \gamma z)(\dots)};$$

viz. this is

$$\Omega\{dx, dy, dz, 2dydz, 2dzdx, 2dxdy\}(\alpha x + \beta y + \gamma z)($$

Here, on the right-hand side, the coefficient of dx is

$$\begin{aligned} &= \alpha(a\xi + h\eta + g\zeta)(\alpha x + \beta y + \gamma z) - (ax + hy + gz)\{\alpha(\alpha\xi + \beta\eta + \gamma\zeta) - \Omega\xi\} \\ &= \alpha\Omega\xi(\alpha x + \beta y + \gamma z), \end{aligned}$$

which is right; and similarly, the coefficients of dy and dz have the same values on the two sides of the equation respectively.

Aronhold's Quadric Integral deduced from the Affected Theorem. Art. Nos. 94 to 99.

94. Let the fixed curve be a conic, say $f = (a, b, c, f, g, h)(x, y, z)^2 = 0$; and let the variable curve be a line meeting the conic in the points 2 and 4. The affected theorem is

$$\Sigma \frac{\delta\phi_2 d\omega}{012} = \frac{\delta 24}{124} + \frac{\delta 24}{234},$$

where (x_1, y_1, z_1) and (x_2, y_2, z_2) are the coordinates of the points 1 and 2 respectively; 12 denotes the constant $(x_1, \xi x_2, y_2, z_2)$, and 012, &c., denote determinants as usual.

The left-hand side is here

$$\frac{1}{12} \left(\frac{\delta \phi_2 d\omega_2}{\delta_4} + \frac{\delta \phi_4 d\omega_4}{\delta_4} \right),$$

on the right-hand side, δ refers to the variation of the constants of ϕ , that is, of the variations of the points 1 and 2; or we may write $\delta = d_1 + d_2$, where the suffixed d 's are independent, and the equation, being satisfied as all, must be satisfied separately

[book p. 165] in regard to the variations of δ , and in regard to the variations of 4: we must therefore have

$$\frac{d\omega_2}{12} = \frac{d_2 24}{124} + \frac{d_2 34}{234},$$

and the like equation obtained therefrom by the interchange of the numbers 2 and 4.

95. The equation just written down relates to any four points 1, 2, 3, 4 of the conic; and if 3 is to vanish, 0, 2 respectively, the equation is reduced to

$$\frac{1}{12} \frac{d\omega_2}{d_2 24} = \frac{d_2 24}{124} - \frac{d_2 01}{012},$$

which may be verified as follows: the equation is obvious; as to the second, observe that writing therein 0 for 4, it becomes $\frac{d_2 02}{012} = -\frac{d_2 02}{012}$, that is, $0 = 2 \frac{d_2 02}{012}$, an identity. And the formula above obtained are

$$\frac{d\omega_2}{12} = \frac{d_2 02}{012} = \frac{d_2 12}{12}, \quad \text{that is,} \quad \frac{d\omega_2}{12} = \frac{12}{12} \left(\frac{d_2 02}{012} - \frac{d_2 12}{12} \right);$$

the equation in question.

96. We have, as a property of any four points 0, 1, 2, 3 of a conic,

$$\frac{23 \ 01}{12 \ 12} = 0, \quad \text{or say} \quad \frac{23 \ 01 \ 23}{012 \ 12 \ 12} = \text{const.}, \quad \text{that is,} \quad \frac{23 \overline{01}}{12 \ 30} = \frac{01}{12};$$

hence considering as a variable point 1, and differentiating the logarithms,

$$d \log \frac{23 \overline{01}}{12} = \frac{d 23}{23} + \frac{d \overline{01}}{01} - \frac{d 12}{12},$$

and the foregoing equation $12 \frac{d\omega_2}{d_2 24} = d \log \frac{23}{12}$ thus becomes $12 \frac{d\omega_2}{d_2 24} = -d \log \frac{01}{12}$, or restoring u in value of 012,

$$12 \frac{d\omega_2}{d_2 24} = d \log \frac{12}{01}.$$

Taking now $\alpha x + \beta y + \gamma z = 0$ for the equation of the line 012; this meets the conic in the points 1 and 2, the coordinates being to be taken: hence 0, 1 are

$$\alpha, \beta, \gamma = y_1 z_2 - y_2 z_1, \ z_1 x_2 - z_2 x_1, \ x_1 y_2 - x_2 y_1,$$

$$12 = ax_1 x_2 + by_1 y_2 + cz_1 z_2 + f(y_1 z_2 + y_2 z_1) + g(z_1 x_2 + z_2 x_1) + h(x_1 y_2 + x_2 y_1),$$

(the second term being of course = 0), viz.

$$\overline{12} = ax_1 x_2 + by_1 y_2 + cz_1 z_2 + f(y_1 z_2 + y_2 z_1) + g(\dots) + h(\dots),$$

$$= -(a, b, c, f, g, h) (y_1 z_2 - y_2 z_1, z_1 x_2 - z_2 x_1, x_1 y_2 - x_2 y_1)^2 + (\dots).$$

[book p. 166] or say $12 = \Omega$, if $\Omega = -(bc - f^2, \dots)(\alpha, \beta, \gamma)^2$ as before; and the equation thus is

$$\frac{\Omega d\omega}{\alpha x + \beta y + \gamma z} = d \log \frac{(a, \dots)(x_1, y_1, z_1)(x, y, z)}{\alpha x + \beta y + \gamma z};$$

or finally, writing (ξ, η, ζ) instead of (x_1, y_1, z_1) , $12 = \Omega$ becomes the coordinates of one of the intersections of the line $\alpha x + \beta y + \gamma z = 0$ with the conic, the equation becomes

$$\frac{\Omega d\omega}{\alpha x + \beta y + \gamma z} = d \log \frac{(a, \dots)(\xi, \eta, \zeta)(x, y, z)}{\alpha x + \beta y + \gamma z},$$

which is Aronhold's quadric integral.

97. (The foregoing property, which may be also written

$$\frac{23\overline{01}}{012\overline{123}} = \frac{01}{23},$$

is verified very easily in the case of four points 0, 1, 2, 3 of a circle: in fact, calling the differential $\Omega\alpha, \beta, \gamma, \delta$, and the proper distances between the points

$$\overline{20} = \alpha\beta \cdot 2 \sin \frac{1}{2}\alpha, \quad \overline{a01} = 2 \sin \frac{1}{2}\beta \cdot 2 \sin \frac{1}{2}\gamma, \quad \overline{2012} = 4 \sin \frac{1}{2}\delta \cdot \dots$$

and so for the other like expressions; each side of the equation thus reduced to $1 \cdot \sin \frac{1}{2}\delta \cdot 1 \cdot 4 \sin \frac{1}{2}\delta \cdot 4 \sin \frac{1}{2}\beta \cdot 4 \sin \frac{1}{2}\gamma$,

$$01\overline{2} = (1 - x + y \sin \frac{1}{2}\delta + 2) 4 \sin \frac{1}{2}\delta \cdot 1 \cdot 4 \sin \frac{1}{2}\beta,$$

$\sin \overline{20} = (1 + \cos \overline{20}) 2 \sin \frac{1}{2}\gamma \cdot \frac{1}{2}\delta$, $LD = 2D\delta$, $LE = 2E\delta$, and so to the squarer in like manner to require $T = 1 - t$, $\alpha = 1$. We can see at once when $\overline{12} = 0$, viz. when the second is the right one.)

98. The foregoing investigation gives for these points $\alpha, \beta, \gamma, \delta = 1, -\delta$. Hence

$$\frac{d\omega}{\sqrt{1-x^2}} = -\frac{1}{2} d \log(x - iy),$$

viz. this is

$$\sin^{-1} x = -\frac{1}{2i} \log(x - iy) = \frac{1}{2i} \log(x + iy).$$

Fixed Curve a Cubic: the Parametric Points 1, 2 consecutive points on the Curve. Art. Nos. 99 to 106.

99. The major function $(x, y, z)^{n-3}$ is taken to be $= \frac{1^2 \cdot 023 - 12^2 \cdot 013}{12s}$, so that, calling the differential $Q d\omega$, we have

$$Q = \frac{1^2 \cdot 023 - 12^2 \cdot 013}{12s \cdot 012}.$$

I take for convenience the cubic to be $f, = (x^2 + y^2 + z^2)w, = 0$; the coordinates of 1 are (x_1, y_1, z_1) , those of 2 are $(x_1 + dx_1, y_1 + dy_1, z_1 + dz_1)$, so that for $\overline{2}$ we have

$$\frac{x_1 dx_1 + y_1 dy_1 + z_1 dz_1}{w} = \overline{2} \quad \text{or} \quad x_1 dx_1 + y_1 dy_1 + z_1 dz_1 = 0;$$

write 1, 1, 1 for x_1, y_1, z_1 , so that the formulæ become for shortness

$$\overline{012} = x dx_1 + y dy_1 + z dz_1, \quad \overline{12s} = x_1 dx_1 + y_1 dy_1 + z_1 dz_1, \quad \dots$$

100. We have

$$\overline{023} = \begin{vmatrix} x, & y, & z \\ x_1, & y_1, & z_1 \\ x_3, & y_3, & z_3 \end{vmatrix}, \quad \overline{012} = \begin{vmatrix} x, & y, & z \\ x_1, & y_1, & z_1 \\ dx_1, & dy_1, & dz_1 \end{vmatrix} = O(1, 3),$$

say. Moreover

$$\overline{023} = \begin{vmatrix} x, & y, & z \\ x_1, & y_1, & z_1 \\ x_2, & y_2, & z_2 \end{vmatrix} = -(yz_1 - y_1z) + x_1(yz_2 - y_2z) - x(y_1z_2 - y_2z_1) (\&c.),$$

which by definition is $\overline{O23}$.

[book p. 168] as the second term may be written; moreover

$$12s \cdot 023 - 12 \cdot 013 + 12^2 \cdot 012 + 2(2ws + s^2) 012 = wz_1 012 \cdot 012 \cdot 012 \cdot 012,$$

$$12s \cdot 013 = \overline{O2s} \cdot 01 - \overline{O1s} \cdot 02 + 2(2ws + s^2 + s^3) 012 + 2(2ws + s) 012 \cdot 012,$$

and hence

$$1^2 \cdot 023 - 12^2 \cdot 013 = -(yz_1 - y_1z) \overline{O2s} + (yz_2 - y_2z_1) \overline{O1s} - 2(2ws + s) \overline{O1s} \overline{012},$$

or reducing by

$$2(s_1z_2 - s_2z_1 + y_1y_2(z_2 - z_1) + z_1z_2(y_2 - y_1)) = (\partial_y y_1 + \partial_z z_1) 012,$$

this is

$$= -2(s_1\overline{s_2} - s_1z_2) 013 - (\partial_y y_2 + \partial_z z_2) 012,$$

which is of the third order.

101. We may show that each of the terms contains the factor $(ds_1)^2$: we have, in fact,

$$y_1 (ds_1)^2 = \frac{l}{y_1} \cdot 2y_1 dz_1 - 2lz_1 \cdot 2z_1 dy_1 + \frac{l}{y_1} (\dots),$$

$$z_1 (ds_1)^2 = \frac{l}{z_1} \cdot 2z_1 dx_1 - 2lx_1 \cdot 2x_1 dz_1 + \frac{l}{z_1} (\dots),$$

$$x_1 (ds_1)^2 = \frac{l}{x_1} \cdot 2x_1 dy_1 - 2ly_1 \cdot 2y_1 dx_1 + \frac{l}{x_1} (\dots),$$

hence, first multiplying by x, y, z and adding, we have

$$(xy_1z_1 + yz_1x_1 + zx_1y_1) (ds_1)^2 = \frac{l}{x_1y_1z_1} \{ (xy_1 - 2lx_1) dy_1 - 2lx_1 (yz_1) dx_1 \} - \frac{2l}{x_1y_1z_1} (\dots).$$

But in virtue of $ax_1^2 + by_1^2 + cz_1^2 = 0$; hence the equation is

$$(xy_1z_1 + yz_1x_1 + zx_1y_1) 012 = (a' + b' + c') 012,$$

the required expression for the first term.

102. Again, multiplying by x_1, y_1, z_1 and adding, we have

$$\Sigma x_1y_1z_1 (ds_1)^2 = \frac{l}{x_1y_1z_1} \{ 4y_1z_1 dx_1 - 2l(x_1 + y_1 + z_1) dx_1 + 4z_1x_1 dy_1 + 4x_1y_1 dz_1 \}.$$

[book p. 169] when, in virtue of $x_1^2 + y_1^2 + z_1^2 = 0$, the first line is

$$= -\frac{2}{x_1 y_1 z_1} \{ (3y_1 z_1 + y_1^2 + z_1^2) dx_1 + \dots \},$$

and this again is $= -2(x_1 dx_1 + y_1 dy_1 + z_1 dz_1)$: in fact, we have identically

$$\begin{aligned} & x_1 y_1 z_1 (2y_1 z_1 + 3\partial_y y_1 + \partial_z z_1) + (x_1^2 + y_1^2 + z_1^2) dx_1 + \dots \\ &= (x_1 y_1 z_1 + 2lx_1 + 2ly_1) dx_1 + (\partial_y y_1 + \partial_z z_1) dx_1 + \dots \end{aligned}$$

which, in virtue of $ax_1^2 + by_1^2 + cz_1^2 = 0$ and $a^2 + b^2 + c^2 = 0$, becomes

$$= a(\partial_y y_1 + \partial_z z_1) (dx_1 + dy_1 + dz_1).$$

Hence the equation is

$$\Sigma x_1 y_1 z_1 (ds_1)^2 = 2(ax_1 + by_1 + cz_1) 012,$$

or finally

$$x_1 y_1 z_1 (ds_1)^2 = (ax_1 + by_1 + cz_1) 012,$$

the required expression for the second term.

103. Writing for shortness

$$1^2 \cdot 023 - 12^2 \cdot 013 = \frac{1}{4}(a, b, c, f, g, h)(x, y, z)^2,$$

we have

$$1^2 \cdot 023 - 12^2 \cdot 013 = -\frac{1}{4}(a, b, c, f, g, h)(x_1, y_1, z_1)(012) \cdot (ds_1)^2,$$

and hence dividing by

$$012 \cdot 12s, = 012 \cdot \frac{1}{2l}(dx_1)^2,$$

we have

$$Q = \frac{1^2 \cdot 023 - 12^2 \cdot 013}{012 \cdot 12s} = -\frac{l}{2}(a, b, c, f, g, h)(x_1, y_1, z_1)(0613, \dots).$$

But this can be further reduced: its numerator, multiplied by 3, is

$$= -(xy_1 z_1 + 2lx_1) \{x_1 \dots\} + \frac{x_1^2}{x_1 y_1 z_1} \{2yz_1 (2ly_1 z_1) + \dots\},$$

which is

$$= y \{x_1 (y_1 - z_1) - 2(x_1^2 - 2ly_1 z_1)\} + z \{(\dots)\} \frac{1}{x_1 y_1 z_1} (\dots),$$

where $a_1(y^2 - z^2)$, $y_1(z^2 - x^2)$, $z_1(x^2 - y^2)$ are the coefficients of the tangential of the point 1 in regard to the cubic, etc. in usual notation; and we have thus

$$Q = \frac{12 \cdot 023 - 12^2 \cdot 013}{012 \cdot 123} = \frac{12 \cdot 613}{012 \cdot 123}$$

where $0\bar{1}.0$ is the equation of the tangent of the line 012; this meets the conic in the points 1, $\bar{1}$ regard to the cubic, where $\bar{1}$ is what may be called the harmonic of the original.

C. XII.

22

[book p. 170] **104.** The identity just referred to is proved very easily. Computing on each side the coefficient of yz , $-yx$, the factor a , divides out and we ought to have

$$-(xy_1 z_1 + 2lx_1 + yz_1 y_1) + (x_1 z_1 + 2lz_1)(y_1^3 - z_1^2) x z_1 = (y_1^2 - z_1^2) a x_1,$$

that is,

$$(yz_1^3 - 2lz_1^3y_1)(y_1^2 - z_1^2) = ax_1.$$

and, in fact, from $x_1^2 = ax_1$, we have for a an integral $\int \frac{dx}{a}$; and the equation is thus verified.

105. The proof has been given in the present case, $f = ax_1^2 + by_1^2 + cz_1^2$, where the cubic is general, the equation is similarly the coefficients of ax_1 , ay_1 , az_1 , equal as the cubic, and the equation is thus verified.

106. We have

$$\overline{028} + \overline{001} + \overline{012} = \overline{12s},$$

or writing this in the form

$$\overline{012} - \overline{[12]} + \overline{021} + \overline{012} = 0;$$

suppose 2 is here the consecutive point $1 + d1$; then

$$\overline{012} - \overline{[12]} + \overline{021} + \overline{012} = 0$$

becomes $= 02(1d1)$; we have also

$$\overline{021} + \overline{012} = -01d12,$$

and the result is

$$-\overline{012} \cdot d\overline{12} = 0,$$

that is, $\overline{012}[\overline{12} + \overline{12} + d61] + d61y012 = 0$, is

$$= \frac{1}{0!} \frac{1}{12!} (d^2d^2 - d^2d^2) \cdots (d^2 - d^2);$$

in the differential of $\overline{012}$ in regard to the parametric point 1 is $\overline{012}_2$, the symbol for the case where the parametric line is the tangent at 1.

Formulæ Relating to the Tangents from the Vertices.

121. I assume some formulæ relating to the tangents to the curve from the vertices A, B, C respectively. We have from each vertex four tangents: say $\rho = 0, \sigma = 0$, symmetrically situate in regard to the axis, and $\sigma = 0, \sigma = 0$, symmetrically situate in regard to the axis: the four tangents from the vertex A are accordingly equations $\rho = 0, \sigma = 0$, denoted by $\rho = 0, \sigma = 0, \rho = 0, \sigma = 0$, the tangents at $\sigma = 0, \rho = 0$, the tangents at $\sigma = 0, \sigma = 0$.

[book p. 171] *Fixed Curve a Cubic: the Parametric Points 1, 2 corresponding points.* Art. Nos. 107 to 110.

107. The parametric points 1, 2 are taken to be corresponding points, that is, such that the tangents at these points meet at a point of the cubic. We may take, in the very particular case considered, the cubic $y^2 = 1 - x \cdot 1 - k^2x$. The four points 1, 1', 2', 2 are then such that the lines 11' and 22', viz. the lines $\tau = 0$ and $\tau' = 0$, meet at a point 7 on the cubic, T being a properly determined linear function of (x, y, z) ; it is a properly determined linear function of (x, y, z) such that the tangents from T to the cubic, that is the line $T = 0$ being any line through the two points 1, 2 meet in a point 9 as above; viz. if we write

$$f = yz^2 + 2lxyz + (x^2 + 2ay + bz)xz = 0$$

and writing

$$d\omega = \frac{x dy - y dx}{\frac{df}{dz}},$$

which, from the equation of the curve written in the form

$$z(yx + 2ay + bz)x + (1 + 2ay + bz)xy + \xi z(xz + by) + lxyz = 0,$$

we have

$$\frac{df}{dz} = \Sigma(xz + by + cz),$$

or say

$$x(xy + by + cz) + xyz = -2lyz\left(\frac{l}{x}\right),$$

becomes

$$= \frac{-2lyz}{\xi},$$

and we have here

$$d\omega = \frac{\xi(dy - ydx)}{-2lyz}, \quad = -\frac{1}{2l}\xi \frac{dx}{(dy)},$$

where $\xi' dx + 6y + lz$. To find the meaning of ξ , observe that the line $x = My = 0$ tangents at the points 1 and 2, which is the case in which the case where, $a = My - 6$ are equal $T = 9$; whence in the case where the tangents at the point 1 and 2, the lines $\overline{1y} = \overline{12}$ and $\overline{6y} = \overline{12}$ meet at a point T on the cubic, T being a properly determined function $\overline{T}$ of (x, y, z) , this line will be a tangent if

$$(yM + \xi)(M^2 + h) = 0,$$

this line will be a tangent at

$$(yM + \xi)(M^2 + \lambda) - 0;$$

22-2

[book p. 172] and we then have the point of contact $(\lambda M + \xi y + \lambda = 0)$, and writing this in the form

$$\xi xy + 2ay + cz = 0;$$

we see that the equation $x = My + \xi z = 0$, the curve $\xi = \xi z + cy + Aps = 0$. Moreover, from the equation of the curve written in the foregoing form

$$\xi(\xi x + 2ay + cy)x + 2yz = 0;$$

it appears that the lines $x = My + \xi z = 0$, $x = My + cz = 0$, viz. the curve, that the line through the two parametric points 1, 2 meets the curve at a point $\overline{T}$ on this it is essential it is essential it is essential.

108. The equation $\xi = 1$ becomes $= ds$, in fact,

$$\begin{aligned} 2. \overline{12} &= (a, b, c)(x_1, y_1, z_1)(x_2, y_2, z_2), \quad (x_1, y_1, z_1) = (1, 0, 0), \\ &= \frac{1}{l} c \xi_2, \\ &= \frac{1}{l} c (xy_2 + \xi_2 x_1 + 2cy_2 + \dots), \quad (x_2, y_2, z_2) = (1, 0, 0), \\ &= b. \end{aligned}$$

We have thus $\xi =$ satisfying the required condition for the major function: and the differential $Q d\omega$ may therefore be taken to be $= \frac{a}{ay}$, that is, we have

$$Q d\omega = \frac{x}{ay} \left(\frac{dx}{ay} \right).$$

The affected theorem then becomes

$$2 \frac{Q \cdot d\omega}{012} = \frac{d\phi_1}{\phi_1} + \frac{d\phi_2}{\phi_2},$$

Integrating each side, and assuming that the superior limits are given by a line ϕ which cuts the cubic in the points 4, 5, 6 and the inferior limits by a line ϕ' cutting it at the points a, β, γ , we have

$$\log \frac{x_1 y_1 z_1 z_1}{x_2 y_2 z_2 y_2} = 2 \log \frac{\phi_1}{\phi'_1},$$

that is,

$$\frac{x_1 y_2 z_3 z_4}{x_4 y_5 z_6 y_2} = \frac{(\phi_1)^2}{(\phi'_1)^2},$$

where ϕ_1, ϕ_2, ϕ_3 denote the values of the linear function ϕ at the points 1 and 2 respectively. We have a cubic cut by the lines ϕ, ψ, x, y in the points 4, 5, 6, 1, $a, \beta, \gamma, 2$, and we have first the points where in course we have ϕ, y replaced by their values, and the formula for the integral of the points (α, β) is

$$\left(\frac{xv}{a}\right) \left(\frac{\phi_1 t}{\phi'_1}\right) \left(\frac{\phi_2 t}{\phi'_2}\right) = \left(\frac{\phi_1 t}{\phi'_1}\right).$$

[book p. 173] that is,

$$\frac{x_1 \bar{y}_2 z_4 z_5 z_6}{x_2 \bar{y}_2 z_a z_\beta z_\gamma} = \frac{\phi_1 \phi_2 \phi_3 \phi_4 \phi_5 \phi_6}{\phi'_1 \phi'_2 \phi'_3 \phi_a \phi_\beta \phi_\gamma},$$

which, dividing out the $\phi_1 \phi_2$, and writing 1, 2 in place of $1', 2'$, becomes

$$= \frac{(\phi_1 \phi_2)^2}{(\phi'_1 \phi'_2)^2},$$

agreeing with the result just obtained.

Aronhold's Cubic Transformation. Art. Nos. 111 to 119.

111. This was obtained in the paper "*Algebraische Reduktion des Integrals $\int F(x, y) dx$, u.s.w.*," Berl. Monatsb., April, 1861, pp. 462–468. I give in the first place the analytical results, independently of the general theory, with the values for the canonical form $f, = \frac{1}{4}(x^2 + y^2 + z^2 + 6lxyz), = 0$, of the cubic. We have

$$T = \text{auxic invariant, } = 1 - 20l^3 - 8l^6,$$

$$S = \text{quartic } (\text{Aronhold's } = 4(l - l^4),$$

$$R = \text{discriminant, } = (1 + 8l^3)^4,$$

$$P = 9b^2 l^2 = [-30l^2 + (1 + 20l^3 + 8l^6)] a + \&c.,$$

$$Q = f y l^2 = (a^3 + 2lc) y x + \&c.,$$

$$B = f y i^2 = (a^3 + 22lac) y x + \&c.,$$

$$C = f x^2 y = (a^3 + 23) y x + \&c.,$$

$$D = f x^2 z = a^2 + yx + x^2 + 6lxyx.$$

where $a, b, c \equiv (P - x)^3, \bar{S}(x^3 - x^3), \bar{x}(z^2 - x^3)$.

Then we have

$$27Q^2 + 54P^2 = (1 - lP)^3 QP = R^2(8C^2 - 8RD),$$

viz. this equation, where each side is a quartic function $(x, y, z)^2$, is an identity when (a, B, c) are connected by the equation, $f a^3 = a^3 + b^3 + 6lab c = 0$; and further,

$$Q4P^2 - PdQ - Rfz_2 = R^2(R^2 - 7R)fz_2 = R^2(R^2 - 7R)fz_2.$$

Hence writing

$$\Lambda = \frac{6loRQ}{(BC)^2} - \frac{2P}{Q} - \frac{Q^3}{Q^3} = \frac{1 - 30W^2 + (1 + 27)Wlc + \&c.}{(b^3 + 22l)a + \&c.},$$

we have

$$Q^*(\Lambda - 3\Lambda + 27) = R^2(C^2 - 8RD).$$

and

$$Q^* d\Lambda_a = 2(QQ^2 - PdQ) = -2R^2(c\sqrt[3]{(cd_a - dd_a)} + \&c.).$$

[book p. 174] **112.** Supposing now that (x, y, z) are the coordinates of a point on the cubic, that $D = 0$; and taking the square root of each side of the first equation, we may write

$$Q^4 = 3RR - 3RS A^2 = -2(P^2 + (yI + dy + \&c.)),$$

We have

$$d\omega = \frac{ax dy + dy dcx}{x^2 + 2lcy} = \frac{a dx - x dz}{a + 2lcy} = \frac{a dy - y dx}{x^3 + 2lcy},$$

whence

$$d\omega = \frac{x(y dx - x dy)}{x^3 + 2lcy},$$

and we consequently have

$$\frac{d\omega}{\sqrt{x^3 - 3x^3 - LR^2\sqrt{x}}} = \frac{1}{\sqrt{R}} \frac{dy}{y^2 - 3y^2 - LR},$$

which, if 128, bP are put $= p, \bar{p}$, respectively, takes the Weierstrassian form

$$\frac{d\omega}{\sqrt{x^3 - px - \bar{p}}} = \frac{1}{\sqrt{R}} \frac{dy}{\sqrt{y^3 - py - \bar{p}}}.$$

The conclusion is that for the cubic curve, taking Λ a quotient of two linear functions of (x, y, z) , the differential $d\omega$ is transformed into $d\Lambda$ a square root of a cubic function of Λ : viz. we have from a form of differential $d\omega$, which belongs to the ordinary theory of elliptic functions, and which has been adopted by Weierstrass as a canonical form.

113. The transformation depends on the arbitrary point (a, β, γ) of the cubic: the point (a, β, γ) is the tangential of this point, viz. the point of intersection of the tangent at (a, β, γ) with the cubic: we can from (a, b, c) draw three tangents to the cubic, viz. the tangent at (a, b, γ) and three other tangents: the equations of the four tangents being $\frac{4P}{Q} - \frac{8P}{f}y = a, \lambda, \lambda, \lambda$, respectively; where $\lambda, \lambda_1, \lambda_2$ are the roots of the equation $\lambda^3 - 3R\lambda - 12 = 0$.

Suppose for a moment that $(\bar{a}, \bar{\beta}, \bar{\gamma})$ is a point on the cubic curve, and write $A = a' + B y + \bar{6}^2$ Such. We have

$$A'D^2 + 4A'C^3 + 6PD^2 - 3DPC^2 - 6A'LCD = 0,$$

for the equation of the six tangents which we know drawn from the point (a, β, γ) to the cubic: when $(\bar{a}, \bar{\beta}, \bar{\gamma})$ is a point on the cubic, $A = 0$, and the equation reduces itself to a quartic

function, $f\bar{a} = a' + \beta + \gamma + 6lab\gamma, = 0$; and the equation $\frac{4P}{Q} - \frac{8P}{f} = 0$, throwing out the factor B , is an identity.

The last-mentioned expression for the $d\Lambda$ depends on the points A, B, C, D as the four tangents to the cubic from the points $-3K, 1, 2K, 1, 6, 27K y \rightarrow b$, those points are the four tangents from (a, β, γ) to the cubic.

Fixed Curve the Quartic $y^2 = (1 - x^2)(1 - k^2x^2)$. Art. Nos. 145 to 145.

145. This is a curve having a uniquely twin to inflexion: there are at this curve being. The points a, β, β' are double points. The equation of the curve are equal $A = a^2, K = a^2$ resolved into seven points 1, 2, 3, 4, 5, 6, 7, with the order of points be $y^2 = (1 - x^2)(1 - k^2x^2), = 0$; and the cubic forms a circle and is involves more terms in the more concrete fashion of 1 and 2. We have $a, b'a = (1 + a)c$; and $b'a = -a'(1 + a^2)c$, those of the four tangents from $(0, 0, 0)$ as expressed by $\frac{6}{4 - x_0}$ are equal $\frac{6}{4 - x_0}$.

[book p. 175] The foregoing expression for $Q dP = P dQ$, viz. the equations of the four tangents at the points of contact thus are obtained, where the dynamical conditions that Aronhold obtained his results.

114. The foregoing expression for $Q dP - P dQ$, viz.

$$Q dP - P dQ = (Aa + By + Cz)(Adx + B dy + C dz) + Aa da + Bb db + Cc dc - (BC' - CB' da) \cdots,$$

may be verified without difficulty. Writing for a moment

$$Q dP - P dQ = (Aa + By + Cz)(\Lambda dx + 3\Lambda dy + Adx) - (Add a + Bdy db + S dc) - (BC' - CB')(La + L db + L da) \cdots,$$

we have

$$\begin{aligned} RX &= -CR + a[-2yQP - (1 + 20l^3)X] - (1 - 5l)(1 + 20X^3 + 5l^2)aY - \&c., \\ &= -(1 + 30l^3)(a^3 + 2lac)a\dot{x} + (1 - 5l)(1 + 5l^2(1 + 8l^3))aYY - \&c., \\ &= -8l^3(1 + 20l^3)(a^3 + 2lc)a\dot{x}, \\ &= -16l(a^3 + 2lac)a\dot{x}, \end{aligned}$$

which prove the theorem.

115. I content myself with a partial verification of the identity

$$2TQ^4 + 6T^2Q^2 - 9T^4P^2 - 4T^4 - (1 + 8L^3)^4(8C^2 - 8RD).$$

Writing herein $a, b, c = 1, 0, 0$, we have $D = 0$, and the equation becomes

$$2TQ^4 + 6T^2Q^2 - 9T^4P^2 - 4T^4 = (1 + 8L^3)^4(8C^2 - 8RD);$$

where now

$$Q = a - 2l(a + \beta - 8l\gamma), \quad P = (a - \beta) - 5l^2(a + \beta - 5l\gamma) + (1 + 25\gamma^2);$$

which, putting therein $a^2 + a^2 + \beta^3 = 6la\beta\gamma$, becomes

$$= -(a - \beta)(a + B - 5l\gamma),$$

Hence writing

$$X = a + \beta - 5l\gamma, \quad Y = -5l(a + \beta) - (1 + 25\gamma^2)\gamma,$$

we have

$$Q = P, \quad C = (a - \beta) X, \quad (a - \beta) Y, \quad (a - \beta) X;$$

substituting these values, the factor $(a - \beta)^4$ divides out, and the equation becomes

$$2TX^2 + 6T^2X - 9T^4 + 4T^4 = (1 + 8T^4)^4(8Y - 8X^4);$$

To complete the verification, observe that we have $Y = 8T^4X^2 - (1 + 8T^4)X$, whence

$$-Y = (1 + 8T^4)(YX + 8(PY - X^2)YT X^4)YT^4X^2.$$

[book p. 176] and herein $-y' = a^2 + 8laa\beta_1$, whence

$$-y' + a^2X = -y + (1 - 5T^2)(8a + 8BX) + 8T^2 + 16yYX + 8T^2yX,$$

that is,

$$-(1 + 8T^2)a + X^2 = a^2 + 5T^2y(X) + 8(15)y(XY) - (X^2X)X^2 + 16T^2X^2.$$

Hence the equation to be verified becomes

$$\begin{aligned} 2TX^2 + 6XT^2X &= (1 + 8X^2)(20X^4 - 4X) + 12LX^2X. \\ &= (1 - 5T^2)(2y - 4y_2)(2X^4 + 16X) + (12L^2)Y \\ &\quad - (X + X^3)Z X^3, \\ &= 0X^2 + 8T^2(y'X)X. \end{aligned}$$

which is again the factor X , and which can be thrown out: the equation thus becomes

$$2TX + 6T^2 + 6TX = (1 + 8T^2)(20X^4 - 16) + (12L^2)(X + 4y) = 0.$$

This may be written

$$2TX + 6T^2 + 6TX = (1 + 8T^2)(20X^4 - 16) + (12L^2)(X + 4y) = 0;$$

or, finally, it is

$$2TX + 6T^2 + 6TX - 30T^4 - 8T^4 + 4(1 + 8T^2)(X^2 + 8T^2)(X + 4y) = 0;$$

$+ - a^2 - X - 4y(1 + 8T^2) + 2T(2 + 16T^2)(X^2 + 8T^2)X = 0$, substituting for T, X their values $1 - 200 - 8l$ and $-a + 4l - 5y$ respectively, the coefficients of X^2 and $(1 + 8l^3)$ are separately $= 0$, and the equation is then proved.

116. The foregoing equation $\Lambda = \frac{6loyQ}{(b^3)^2} - \frac{8P}{fy}$, regarding therein Λ as an arbitrary parameter and (x, y, z) as current coordinates, is the equation of an arbitrary line through the point (a, b, c) of the cubic: it meets the cubic in two other points (depending, of course, on the value of Λ); and the equation considered as relating to these points is

[book p. 177] $\frac{Efa}{fy} = \frac{C\Lambda}{\sqrt{P^3 - 3\Lambda S^2 + T}}$. The expressions answer a peculiarly simpler form when (a, β, γ) , instead of being an arbitrary point of the cubic, is a point of inflexion on the cubic; and it is easy to see *a priori* why this is so: in fact, if we assume

$$x : y : z = a + \lambda M, \quad \beta + \lambda N, \quad \gamma + \lambda R;$$

where a, β, γ are linear functions of Λ and the modulus a , the inflexion point of the cubic from which we proceed; observe also, that depending on a, β, γ , take an integral, by elimination, such a proper inflexion at the point of inflexion, we have

$$x : y : z = a + \Lambda M, \beta + \Lambda N, \gamma + \Lambda R.$$

where, λ, ν, ρ are constants! And then with these values of Λ, M , and for elimination, Λ, ν, ρ are also linear function of the equation of Λ , viz. $(\lambda)(\beta - \lambda P X) c) \rho = X, y(\lambda - \beta W)$.

117. To work this out, we start from the equation

$$\Lambda = \frac{6loyQ}{(BC)^2} - \frac{2P}{fy}, \quad fa = (1 + 8l^3)a, \quad B = 2aa^2, \quad C = 12lc^2a,$$

hereon $Aa + By + Cz = 0$ is the equation of the line through the point $a, b, c = 0, 0, a$; or as before, $a = 0$. We can the same equation written, $Ab + By + 0, Cya$.

We have $a, b, c = 0, 0, a$; $B = a, y, z = 4a, b, c, ax^3, C = 12lc^2 = 12y, LD = 6lz, \xi = D$; in ξ at $(0, 0, a)$, the cubic function reduces to a in the form

$$Ax + By + Cz = -4ay^2 + 12lyz + 6lyz = (-4a + 12ly + 6lz)y;$$

from which we as yet unknown coefficient LR is to be obtained. This is most easily effected by assuming $a, y, a, 0$, values which give Λ obtained $-4a - 1$; whence

$$C' = a^4 - 6la(yxy + y + (1 + 8l^3)Ly'(y'a^4) - 6lxy sc) = 2L(y'),$$

$\delta C = 2L(y(yx + yy) + 6ly a(yx + yx) - 24l(ya)c)$. from which on yet unknown coefficient L is to be obtained. This most easily effected by assuming $a, y, a, 0$, values which give $\Lambda = a$ obtained. We need easily obtain by writing $y = a^2, 0, 0$, which from the previous results so above

$$C' = (1 + 4a^2(1 + 8l^3)aY \cdots ()).$$

the whole equation becomes a quartic in y , satisfied by all, must be satisfied separately

$$C = U + (3l^2(C + 8l^3)L((PA^2 + 3(LA^2)L(CPB + A^2)(CL(A^2)A^2 + 121\rho A^2).$$

[book p. 178] where for C is to be substituted the value $LX + 2RAS$. We may also reduce by reducing also the right-hand side, the whole equation becomes

$$C' = L'(x' + By + 6lG\bar{a})(C + LX);$$

which after reduction is found to contain only the factor a , and omitting this factor and reducing also the right-hand side, the whole equation becomes

$$\begin{aligned} &L(x^3 + Dy + 6ly) + L^2A^2 = -3yC, \\ &+LM(By + 34la)B + LM(C + B,)B^2(LX)(LX) \\ &+(LX)(Q + 26la^2)(C + 20) \\ &+LB(LX + 6lB + 9(LL)B + 4l(LL(CL)L^2 + 0)4 \\ &+(BL^2Q + LX^2B + 2L)M(P) + 2LLM + 0)P \\ &+4(LB - 4lQ)(B + 3l^2)Q + LM(P)P + 12X^2 + 0)P \\ &+1(2 + LA^2)P; \end{aligned}$$

omitting from these the factor C , on putting therein the values of C , D , the equation contains a^2 as a factor; and dividing also throughout by aS , the equation becomes a quadric function in Λ and the equation thus contains Λ , LX , LX as the coefficients of the resulting quadric.

We find very easily that the differentiating, always by $a + 6y + 6lyz = 0$;

$$\frac{1}{2}(2P - 9R) = ly^2 + 6lyzX' + 2L(X + BX);$$

But we have

$$(L = (LL) + 6lLX + RY(C LX) + 2lL(RL)X)$$

the equation in (x, y) is

$$By = -3(2x + 9y)Y;$$

or

$$y^2 + 9y = d^2 \sqrt{X}.$$

We had evident difficulty, that taking always $y = 4 + 6y(LXY)X = 0$, the factor which divides out being in each of these cases

$$\sqrt{2} = \sqrt{x}(1 + a),$$

[book p. 179] which is

$$= (LL - M)(LA^2)LLM + B(L LX)XL - LLXL - LLXL;$$

But we have

$$(L - M)(z + LLX)LX + (L - M)YXY(XYXL)XL - LL,$$

showing that the solution involves the factor $\sqrt{LX^3 + 5LX^2 - X^2}$.

$$\frac{1}{2}(QL - LXL)XL = (1 + 8L^3)XL - 2L(L + l + 2L^2)L(L^2YL^2).$$

showing that the equation in (z, y) is

$$A = \frac{-6L^2 - L(L^2 + 2T^2 + 8X^2)y}{x + y - 5Lz};$$

the equation in (x, y) is, in fact,

$$(B + 8L^3)y^2 + (1L + Lc)y + (-6L^2 + L(L^2 + 2L^2))y = 0,$$

or, as this may be written,

$$(L + 8L^3)x_a + 2LYy_a + 8L^2X_a + 8X_a^2X_a + 4l(LL(L, M))y = 0;$$

viz. we thus have

$$\sqrt{1 - 25X^2} = (1 + 25L)a + \sqrt{2L + 8L^2 - 6T};$$

or, retaining M in the relation

$$y = \sqrt{2}(a + 5L)a + 5L\sqrt{1 - 25X^2 - 8L^4 - 8X^2};$$

$x = \sqrt{2}(a + 5L)a + 5L\sqrt{1 - 25X^2 - 8L^4 - 8X^2}$; viz. we thus have

$$x = \sqrt{2}(1 + 8L^4 - 8L^3 - 8L^4 - 8L^4);$$

where also

$$x = \sqrt{2}(2 + a),$$

these values satisfying obviously

$$(a + 6y)(1 - L^2)^2 = (a + 6y)yL^4 - 27T.$$

119. Starting from these values we, in fact, easily obtain

$$\frac{a dy + y dx}{a^4 + y^4 + a^2} = \frac{-\sqrt{x} dx}{\sqrt{x - 8X^4}};$$

$$2y d\dot{x} + (1 - 6L^3)(d\dot{x}) = -(12 + 12L^3 + 8x^3) - 4(-72 - 8L)a^2,$$

$a^2 - 25y_4 = 2(11x_a)$, and hence

$$d\omega = \frac{x dy + y dx}{a^4 + y^4 + a^2} = \frac{1\sqrt{x} dx}{\sqrt{x - 8X^4 - 8LX}}.$$

The same result might of course have been obtained from the values of a , x , or y , a : the factor which divides out being in each of these cases respectively.

23-2

[book p. 180] *The Cubic* $y^2 = x \cdot (1 - x) \cdot (1 - k^2x)$. Art. Nos. 120 to 130 (*several sub-headings*).

120. The curve is a semi-cubical parabola, symmetrical in regard to the axis of x ; and if, as usual, k^2 is to be taken to be real, positive and less than 1, the curve consists of an oval, and an infinite portion which may be called the sagitta. (See Figure.)

The equation is satisfied by

$$x = \operatorname{sn}^2 u,$$

$$y = \operatorname{sn} u \operatorname{cn} u \operatorname{dn} u.$$

Observe that the periods for these combinations of the elliptic functions are $2K$, $2iK'$; in fact, we have

$$\operatorname{sn}(u + 2K) = -\operatorname{sn} u, \operatorname{sn}(u + 2iK') = \operatorname{sn} u,$$

$$\operatorname{cn} = -\operatorname{cn} u, \operatorname{cn} = -\operatorname{cn} u,$$

$$\operatorname{dn} = \operatorname{dn} u, \operatorname{dn} = -\operatorname{dn} u.$$

whence the sn^2 and the $\operatorname{sn} u \operatorname{cn} u \operatorname{dn} u$ both unaltered by the change of u into $u + 2K$ or $u + 2iK'$. Hence to a given u as regards sn may be equally well taken to be a determinate value (for sn^2 is a determinate value $\operatorname{sn} u$), we may, for instance, attribute to u any one of the four points belonging to those values of u respectively; thus, $\{u, u + K, u + 2iK', u + K + iK'\}$.

121. If, to fix the ideas, we select for each point of the curve one of the geometrical arguments, viz., that arising from a positive value of the radical, then taking 1, $1'$, 2, $2'$ as in the figure: for these points the arguments are $u, u + 2K, -u, 2K + u$ respectively.

[book p. 181] infinity $v = 0$, and $u' = -K$; below the axis $v = -v + iK'$, $v = u + iK'$, where $u + 2iK'$ are conjugate; thus $u + 2 = u + 2iK'$. Hence these two arguments are negative integers: more generally, $u' = u + 2(mK + m'iK')$; for the present we may consider only A, A', A, a, a, C ; but the two values AC, a ; but the next have, $aC = u + AK'$, etc.

122. The poor theorem gives for these points u_1, u_2, u_3 a line

$$du_1 + du_2 + du_3 = 0.$$

and thence $u_1 + u_2 + u_3 = C$. The constant C cannot have a determinate value (for if it had assuming the values of u_1, u_2, u_3 are points where, by some interpretation of u , the arguments are points on the cubic; the constant of $u_1 + u_2 + u_3 = AK'$; for the constant $u_1 + u_2 + u_3 = u_1 + u_2$, and substituting in the form $u_1 = u + 2iK'$, $u_1 = -u_2 + iK'$; for points "on the sagitta" ($AC + a = -2iK' - iK'$), and substituting in

$$u_1 + u_2 + u_3 = C \quad \text{viz.} \quad u + u_3 + 2iK' = C,$$

the values are simplifying obviously

$$(K + 1)C = (1 - K_a)(1 - K_a)(1 - K_2)(1 - K_3),$$

and thence

$$a_1 = au_1 - K;$$

the values of P, G are easily found to be

$$F = y_1^2 + 2(y_2 + y_3) - z_1(1 + z_1z_2 + z_2z_3 + z_3z_1) + 2k^2(1 - z_1)(1 - z_2)(1 - z_3),$$

$$G = 2y_1y_2(y_2 + y_3) + 2y_2y_3(y_3 + y_1) + 2y_3y_1(y_1 + y_2) k k k k (\dots).$$

[book p. 182] or, as these may be written,

$$F = a_1 + y_2 + y_3 + a_1 + (a_1^2 - a_1) + (b_1^2 - b_1) + (\dots),$$

$$G = a_1y_1 - \frac{1}{4k}a_1; \quad y_1 = \frac{1}{4} + 4K; \quad a_1 = a_1(1 - K)(1 - K)(1 - K),$$

where of course x, y, a are to be replaced by their values $a, a_1, y_1y_2y_3$, etc. This is, in fact, the ordinary process of finding the third point of intersection of a cubic by a line which meets it in two given points.

Writing $aK - a_1$, and a, b, c, d to denote a_1, b_1, c_1, d respectively, we have

$$\frac{a_1}{a_2a_3a_4} \frac{a_2}{a_2a_3a_4} = \frac{1}{abcd};$$

whence

$$a_1^2 = -\frac{1}{a^2a^2}; \quad a_2a_3a_4a_1 - \frac{a^2a^2a^2}{a^2a^2a^2} - \frac{a^2a^2}{a^2a^2};$$

and the relative equation lookings:

$$(1 - k^2a^2a^2)a^2 + (1 - k^2a^2a^2)k^2(a^2) [(k)] = 0,$$

corresponding to the four tangents which can be drawn from the point u to the cubic.

The Four Tangents from a Point of the Cubic.

123. Suppose that the line is a tangent to the cubic, and let us regard the cubic at the point u , and again meets it at the point u' : then instead of u_1, u_2, u_3 , we have u', u, u' and the relation become $2u + u'C$.

Here a being given, u' is uniquely determined; this gives u' , the unique tangent to the cubic at the point u , viz., to the value $u' = a_1, a_1, u' = a_1$, corresponding to the four tangents which can be drawn from the point u to the cubic.

[book p. 183] The tangents are real for a point of the sagitta, and for such a point we may write $u = iK + v$, where v is real and included between the values $\pm K$; the corresponding values of u' are

$$u' = -2u = -2iK' - 2v = -2v + 2K; \quad \text{for inconsistent to coval,}$$

$$\begin{aligned}
u' &= -\frac{1}{2}u + K = -\frac{1}{2}v - \frac{1}{2}iK' + K, & \text{extramediate to oval,} \\
u' &= -\frac{1}{2}u + iK' = -\frac{1}{2}v + iK' = -\frac{1}{2}v + \frac{1}{2}iK', & \text{intermediate to coval;} \\
u' &= -\frac{1}{2}u + K + iK' = -\frac{1}{2}v + \frac{1}{2}iK' + K, & \text{intermediate to oval.}
\end{aligned}$$

The first and second of these belonging to tangents to the oval, the third and fourth to tangents to the sagitta. We may further distinguish a tangent according as it passes between an oval and not passes between: the writing in the tangents from $K = -2v + 2K$, that is, the tangents from the points of the oval $K = K(K + 1)$; but the tangents from the point $iK' + v$ are conjugate. We may, for distinction, call them 1822 and 1823, etc.

124. We may make a corresponding division of the real lines which meet the curve in three real points: any such line meets the oval twice (and twice or three times) and taking the singular three times; and taking the arguments to be a, u_2, u_3 , for the tangents from the point $iK' + v$, we have

$$\begin{aligned}
\frac{1}{2}(u_1 + u_2 + u_3) &= \frac{1}{2}iK', & \text{for intermediate line meeting oval twice,} \\
\frac{1}{2}(u_1 + u_2 + u_3) &= \frac{1}{2}iK', & \text{extramediate line meeting oval twice,} \\
\frac{1}{2}(u_1 + u_2 + u_3) &= \frac{1}{2}iK', & \text{extramediate line meeting sagitta three times,} \\
\frac{1}{2}(u_1 + u_2 + u_3) &= \frac{1}{2}iK' + K, & \text{intermediate line meeting sagitta three times,}
\end{aligned}$$

above and below the axis respectively.

125. Returning to the tangents, the point $iK' + v$ may be an inflexion: we have then the point of contact of the intermediate tangent to the sagitta coinciding with the point itself, viz. writing $iK' + v$ for u and $-\frac{1}{2}v + \frac{1}{2}iK'$ for u' , this requires $v = \frac{1}{2}iK' + \frac{1}{2}K$ and $-\frac{1}{2}v + \frac{1}{2}iK' = -K$; and for these values of the argument v for u , we have, $iK' + \frac{1}{2}K$, the inflexion above and below the axis respectively.

In the case of an inflexion, $u = u_1$, and we have

$$u' = C - 2u_1 = u_1 + 2u_2(u'_a - u_1),$$

to satisfying the four tangents from the point of inflexion $u = u_1, u = u_1, u = u_1, u = u_1, u = u_1$.

If the line be a tangent, this will be satisfied by $u = u_1$, the condition for this is $2v(P + 2)(P + 1)Q = 0$, and in any of the two other points determined by the equation

$$(2y^2 + 1)GH + g(P + 1)(LH + bL) = 0.$$

this line will be a tangent if

$$(yM + 1\sigma M^2 + 1\sigma) - 0;$$

[book p. 184] giving for the equation

$$3Mw + bC Du + 6RDa - B = 0,$$

or for B, C, D writing their values $1, -(1 + k^2)P, kP$ this is

$$3kw - 4(1 + k^2)a + B^2 = 0;$$

for the a -coordinates of the inflexion. The form $u = -iK + K + iK'$, and $-iK + K + iK'$, give us $-iK = K + iK'$ in $iK + iK' = K(1 + iK')$, and others corresponding terms by the interaction of the numbers 3 and 4.

It is clear that, in addition to a roots; positive or negative, of these roots are real, all positive or all negative; whilst it is also a $-1, -1, -1$, which is the third value. In an inflexion above and the inflexion below the axis being symmetrically situate in regard to the axis.

The Sextactic Points.

128. The vertices A, B, C are each of them a sextactic point: in fact, writing the equation in the form $y^2 = (1 - x^2)(1 - k^2x^2)$, we see at once that the conic $y^2 = (1 - x)(1 - k^2x)$, $= 0$ in regard to coincide and C are real inflexions; and besides these none on the cubic in the points A, A, A, A, C, C .

129. The conics having a vertex at one of these points have each a contact of the second order with the cubic; but inquire how the others may.

Formulæ Relating to the Tangents from the Vertices.

129. I assume some formulæ relating to the tangents to the curve from the vertices A, B, C respectively. We have from each vertex four tangents: say at C , $\rho = 0$, $\rho' = 0$, symmetrically situate in regard to the axis, $\sigma = 0$, $\sigma' = 0$, symmetrically situate in regard to the axis: the four tangents from C are accordingly equations $\rho = 0$, $\sigma = 0$, $\rho' = 0$, $\sigma' = 0$, denoted by $\rho = 0$, $\rho' = 0$.

[book p. 185] eight angles to the axis, and that joining the points of contact of ρ' , σ' is the line $x = w'$ also perpendicular to the axis.

Values ρ , Tangents, Imaginary.

130.

$$\begin{aligned}\rho, \rho_1, \rho_2 &= x - y(1 + i)x, 1 + ix, 0, 1 + k; \\ \sigma, \sigma_1, \sigma_2 &= x + y(1 + i)x, 1 + ix, 0, 1 - k; \\ \rho', \rho'_1, \rho'_2 &= x - y(1 - i)x, 1 - ix, 0, 1 + k; \\ \sigma', \sigma'_1, \sigma'_2 &= x + y(1 - i)x, 1 - ix, 0, 1 - k;\end{aligned}$$

the equation of the curve is

$$f = yp - r(x + y)\rho\rho'; \quad \text{Coefficients of Powers of Coords.}$$

Values ρ , Tangents, Imaginary. Coordinates.

$$\rho, \sigma = x - y - (1 + i)kx, kx - x - (1 - i), \quad x = \frac{1}{i+k}, y = \frac{(1+i)(1-k)}{k};$$

$$\rho', \sigma' = x - y - (1 - i)kx, kx - x - (1 + i), \quad x = \frac{1}{i+k}, y = \frac{(1+i)(1+k)}{k};$$

the equation of the curve is

$$f = y\rho(x + i)\rho\sigma' = y(-x - i)\sigma\sigma'; \quad \text{Coords.}$$

Values σ , Tangents, Real. Contacts.

$$\rho, \sigma = x - y - (1 + k)(1 + i)kx, 1 - x - (1 + i), \quad x = \frac{1}{1+i+k}, y = \frac{(1+k)(1-k)}{1+i};$$

$$\rho', \sigma' = x - y - (1 - k)(1 - i)kx, 1 - x - (1 - i), \quad x = \frac{1}{1-i+k}, y = -\frac{(1+k)(1-k)}{1-i};$$

the equation of the curve is

$$f = y(-1 + k)\rho\rho' = y(-1 - k)\sigma\sigma'; \quad \text{Coords.}$$

130. These functions ρ , σ , &c., considering therein x , y as depending therein on the elliptic function $\operatorname{sn} u \operatorname{cn} u \operatorname{dn} u$, are functions of u : I propose to demonstrate from the values of the constants Nos. 163 of my *Elliptic Functions Memoir*, p. 78; viz. we have

$$\operatorname{sn}(v + \tfrac{1}{2}K) = \frac{1}{\sqrt{k}} \frac{1+k-k(1-k) \operatorname{sn}^2 v}{1+k-k(1+k) \operatorname{sn}^2 v},$$

$$\operatorname{cn}(v + \tfrac{1}{2}K) = \frac{1}{\sqrt{k}} \frac{\operatorname{cn} v + i \operatorname{dn} v}{(1+k')(1-\operatorname{cn} v)}, \quad (\text{vertex C});$$

which is

$$= -\frac{1-y-i(1-x)}{1+y-i(1+k)};$$

and

$$\operatorname{sn}(u + \tfrac{1}{2}K + \tfrac{1}{2}iK') = -i \frac{\operatorname{cn} v - i \operatorname{dn} v}{(1+k')(1-\operatorname{cn} v)}; \quad (\text{vertex A});$$

which is

$$= -\frac{1-y+i(1-k)}{1+y-i(1+k)}.$$

C. XII.

24

[book p. 186] which is

$$= \frac{1-y+i(1-kx)}{1+y-i(1+kx)} = \frac{1}{\sigma};$$

Observe that this is the inverse formula, the inflexion which is not a sextactic point, viz. that is, $z = u(1-x) + i(y+kx) - 1 \cdot u$, $\rho\pi$, etc. But neither of these is constant; and so for any other vertex. This is, in fact, no inflexion, and consequently we have

$$\rho\rho'\sigma\sigma' = \rho^2\sigma'^2,$$

and the identity is shown by some calculation of $\operatorname{sn}(u + \tfrac{1}{2}K) - i\rho$, which can be in like manner shown.

Fixed Curve a Quartic in Space; the Quadri-quadric Curve $y^2 = 1 - x^2$, $z^2 = 1 - k^2x^2$. Art. Nos. 131 to 144.

131. It is assumed that k is real, positive, and less than unity: the curve may be regarded as the intersection of the two cylinders $z^2 = 1 - x^2$, $y^2 = 1 - k^2x^2$, but the conception of the curve as belonging to an elliptic cylinder $\frac{2}{1+k^2} - \frac{y^2}{1+k^2} = 1$, and a circular cylinder, $\frac{x^2}{1+k^2+2k^2z}$, are also adopted by Weierstrass as a canonical form.

132. The foregoing expression for $Q dP - P dQ$ is satisfied by the equations $y = \operatorname{cn} u$, $z = \operatorname{dn} u$, $w = \operatorname{sn} u$, $y = \operatorname{cn} u$, $z = \operatorname{dn} u$, $y = \operatorname{sn} u$.

[book p. 187] five points, and then the generating lines come each of them to coincide with the tangent at A , we have the osculating planes meeting the curve in the points A counting four times. We will, therefore, taking A and A' as opposite vertices of a parallelogram, but A' as one of two parallelograms; but A' , A'' , B' , B'' , B'' being the vertices of a quadrilateral.

132. The equations are satisfied by writing therein

$$x, y, z = \operatorname{sn} u, \operatorname{cn} u, \operatorname{dn} u;$$

the periods are here $4K$, $4iK'$; whence it is easy to see that the points A , A' , B , C have for arguments 0 , $2K$, iK' , K , $K + iK'$, $K - iK'$, K , $K - iK'$, $K + iK'$ which are $4 - 0 \cdot iK + 0$.

Similarly the Reformatted form depends on $u + 2K + iK'$, and so reduce the number is like manner to $K + 1 - 1 + 0$, with that the the form which might has been to taken to be

$$x = u, y = u(K - iK') \&(\text{some equation});$$

$0 = K - iK'$, intermediate. Hence &c. in fact, from the values u for which $a = 0$, viz. $u = iK'$, in this case the values u, z, y are like the case of the conic.

133. Returning to the tangents, the point $iK' + v$ may be an inflexion: we have then the point of contact of the intermediate tangent to the sagitta coinciding with the point itself, viz. writing $iK' + v$ for u and $-\frac{1}{2}v + \frac{1}{2}iK'$ for u' , this requires $v = \frac{1}{2}iK' + \frac{1}{2}K$ and $-\frac{1}{2}v + \frac{1}{2}iK' = -K$. The same reduction will be shown directly. Moreover we have

$$BX^2 - 4lac(1 - 2lac)x = a^2 + 4lac(1 - 2lac)x = 0,$$

$$+B - C(1 + a + 1 + Y)B + L - 5(C - LK + 6Y + 4y + 4;$$

$$+B - LK a + ((B + 4l)a + (a + 1)(b - K) + 0)x.$$

$$+2(c - LK)B = -4y(b - K) + a(b + L)(1 - K)b - L(LK) + 0, + -16Z(b - K)x + LK - 1(b + L)L(LK) + 0(LL + 0X),$$

24-2

[book p. 188] which are the required expressions. If we wish to introduce $u_4 = -u_1 - u_2 - u_3$, the resulting equations give the values $\text{sn } u_4, \text{cn } u_4, \text{dn } u_4$ of the fourth point, and in this way identifiable with the ordinary results.

132. The determinant equation may be written

$$(x_1 - x_2, \frac{1}{y_2} \frac{x_1}{y_1} - \frac{x_2}{y_2}) + (x_2 - x_3, \frac{1}{y_3} \frac{x_2}{y_2} - \frac{x_3}{y_3}) + (x_1 - x_3, \frac{1}{y_2} \frac{x_1}{y_1} - \frac{x_3}{y_3}) + (\dots) = 0,$$

and, in fact, each of the three lines is separately $= 0$. This appears from the following three formulæ:—

$$\begin{aligned} \frac{\text{cn}(u_1 + u_2)}{\text{cn}(u_1 - u_2)} &= \frac{x_2 - x_1}{x_2 + x_1}, & \frac{x_2 - x_1}{x_2 + x_1} &= \frac{y_2 - y_1}{y_2 + y_1}, \\ \frac{\text{sn}(u_1 + u_2)}{\text{sn}(u_1 - u_2)} &= \frac{y_2 - y_1}{y_2 + y_1}, & \frac{x_1}{y_1} - \frac{x_2}{y_2} &= -\frac{2(x_1 - x_2)}{y_1 + y_2}, \\ \frac{\text{dn}(u_1 + u_2)}{\text{dn}(u_1 - u_2)} &= \frac{z_2 - z_1}{z_2 + z_1}, & \frac{1}{y_2} \frac{1}{y_1} - \frac{1}{y_2} &= \frac{1}{2}(z_1 - z_2), \end{aligned}$$

which are themselves at once deducible from the formulæ

$$\text{sn}(u + v) + \text{sn}(u - v) = \frac{2 \text{sn } u \text{ cn } v \text{ dn } v}{1 - k^2 \text{sn}^2 u \text{sn}^2 v}, \quad \text{etc.}$$

In fact, the numerators of $\text{cn}(u_1 + u_2) + \text{cn}(u_1 - u_2)$, $\text{sn}(u_1 + u_2) - \text{sn}(u_1 - u_2)$ thus become $= 2(x_1 - x_2)(1 - k^2 x_1 x_2)$, $(2(x_1 - x_2)(1 - k^2 x_1 x_2))y_1 y_2 z_1 z_2$, respectively; so that, taking the numerator of $\text{cn}(u_1 + u_2) + \text{cn}(u_1 - u_2)$ as unity, we obtain by division the formulæ in question.

134. Then above formulæ, putting therein $u_1 = u_3, = 2u$, become, on writing the functions of $2u$, hence

$$\frac{1}{2} \frac{dx}{dx} = -\frac{x_1 - x_2}{y_3 - y_1}, \quad \text{giving} \quad -Dx_1 = \frac{2(x_2 - x_1)y_1}{1 - x_1^2 + (x_1 - x_2)(x_2 + 1)} = \frac{2(x_2 - x_1)y_1}{x_2 - x_1};$$

$$\frac{Dy_2}{2} = \frac{y_2 - y_1}{x_2 - x_1}, \quad = \frac{y_1 y_2 + y_2 y_1 - (1 + k^2)}{x_2 - x_1};$$

$$\frac{Dy_2}{2dy} = \frac{x_2 - x_1}{x_2 - x_1} = \frac{(1 + k^2)(x_1 - x_2)(1 + x_1)}{(x_1 + x_2)(x_2 - x_1)};$$

equations which must of course give each of them the same value for u' : the equations belong to the relation $u = u_1 + u_2$, $u_3 - u_2$, since $(x_2 - x_2)$ are conjugates of (x_1, y_1, x_1) and (x_2, y_2, z_2) respectively.

134. Then above formulæ, putting therein $u_3 = u_4 = -2u$, and the corresponding functions of $2u$, become.

[book p. 189] Write

$$a, \beta, \gamma, \delta, p, q = u_1 - u_2, u_2 - u_3, u_3 - u_1, u_1 + u_2 + u_3, u_1 - u_3, u_2 - u_3;$$

$a', \beta', \gamma', \delta' = u + 1$, writing $\xi = u_1$, it becomes, in place of a, β, γ, δ , in any of $a, \beta, \gamma, \delta, \beta, \beta, \gamma, \delta, \gamma, \gamma, \delta, \delta, \delta$. The determinant equation is satisfied by these formulæ:—

$$a^2 + \beta^2 + \gamma^2 + \delta^2 = -\frac{1}{12k}(L^2 + 2y) + L^2P^2 = 0,$$

$$ax + by + cz + d = a + \beta + \gamma + \delta(d - L) + 8(L^2 + 2 + 4y), \quad au + ax + by + bx + dx = 0;$$

By what precedes, it appears that not only is this so but that we have separately

$$au^2 = a + \beta - 5l\gamma, \quad bu^2 - 5l(a + B) - (1 + 25\gamma^2)\gamma + (1 + 2L^2)P^2 = 0, \quad cu^2 = 0,$$

then not only the original equation, but each of the three equations, holds good.

The Cubic Curve $xy = 2y + (4 + 1)^2z$; $xy - 1 - \sqrt{1 - x^2}\sqrt{1 - k^2}$. Art. Nos. 136 and 137.

135. Writing the equation in the form

$$(xy - 1)^4 = x^4(1 - k^2x^2), \quad \text{or} \quad xy - 1 = \sqrt{1 - x^2}\sqrt{1 - k^2x^2},$$

we see that the general form is as shown in the figure: the end portions of the curve

[book p. 190] curve lie between the values $x = -\infty, -\frac{1}{k}, \text{ and } \frac{1}{k}, \infty$. The curve may be made to depend on an elliptic function by means of the equation; we may here write

$$x = \text{sn } u = \frac{\text{sn}(u + \frac{1}{2}K) \text{ dn}(u + \frac{1}{2}K)}{\text{cn}(u + \frac{1}{2}K)}, \quad \text{say} \quad x = \frac{\text{sn } v \text{ dn } v}{\text{cn } v},$$

where the function sn, cn, dn belong to the second, that is, to the second equation; and herein for u writing $v + \frac{1}{2}K$, and for k writing $k' = \sqrt{1 - k^2}$, we have, of course, the formulæ given.

But from the formulæ, p. 63, of my *Elliptic Functions*,

$$\text{sn}(2v, k) \text{ cn}(2v, k) = \frac{\text{sn}(v, k') \text{ dn}(v, k')}{\text{cn}(v, k')}, \quad \text{dn}(2v, k) = \frac{\text{dn}(v, k')}{\text{cn}(v, k')},$$

and herein for v writing $\frac{1}{2}v$ and for k writing $k' = \sqrt{1 - k^2}$, we have, of course, the formulæ given.

Fixed Curve the Cubic $y^2 = x(1 - x)(1 - k^2x)$. Art. Nos. 138 to 142.

138. It was shown, No. 63, how for the affected theorem, when the fixed curve is a cubic, the form of differential is the multiplied by

$$\frac{012}{012} = \left[\int_a^x (d\omega)_3 (d\omega)_1 (d\omega)_2 \dots \right],$$

the last term being the properly determined constant K , attached to the variable term $\overline{012}$, in order to obtain a standard form of integral. The object of the present articles is to show what these formulæ become for the before-mentioned form of cubic curve $y^2 = x(1-x)(1-k^2x)$, which is most directly connected with elliptic functions: and to exhibit the connexion of the formulæ with the elliptic integrals of the second and the third kinds respectively.

139. We have, in general,

$$\overline{012}, \overline{0\overline{12}} = \left| \begin{array}{c} (1+k')x_1y_1, -(x_1+x_2+y_2)y_1 \\ \xi(x_1+1) + (\xi+1)x_2, (c_0, c_1, x_1) \\ +y(x_1)x_2, \end{array} \right|, \quad \left| \begin{array}{ccc} x, & y, & 1 \\ x_1, & y_1, & 1 \\ x_2, & y_2, & 1 \end{array} \right|$$

[book p. 191] where the sn, cn, dn refer to the modulus k' , a, z, d denote $\text{sn } v, \text{cn } v, \text{dn } v$, k modulus k .

But from the formulæ, p. 63, of my *Elliptic Functions*,

$$\text{sn}(2v, k) \text{cn}(2v, k) = \frac{\text{sn}(v, k') \text{dn}(v, k')}{\text{cn}(v, k')}, \quad \text{dn}(2v, k) = \frac{1}{\text{cn}(v, k')},$$

and herein for v writing $\frac{1}{2}v$ and for k writing $k' = \sqrt{1-k^2}$, we have, of course, the formulæ given.

Fixed Curve the Cubic $y^2 = x(1-x)(1-k^2x)$. Art. Nos. 138 to 142.

138. It was shown, No. 63, how for the affected theorem, when the fixed curve is a cubic, the form of differential is the multiplied by $\overline{012}/\overline{0\overline{12}}$.

139. We have

$$\text{sn}(v + \frac{1}{2}K) = \frac{\text{cn } v}{\text{cn}(v + \frac{1}{2}K)}, \quad \text{dn}(v + \frac{1}{2}K) = \frac{1+k}{1+k} \frac{\text{cn } v}{\text{cn}(v + \frac{1}{2}K)},$$

$$\text{cn}(v + \frac{1}{2}K) = \frac{1}{1+k} [1+k + (1-k) \text{cn } v], \quad \text{dn}(v + \frac{1}{2}K) = \frac{1-k^2}{1+k} [1+k - (1-k) \text{cn } v];$$

and the formula above clear obtained are

$$\text{cn}(1+k, k) = (1+k)(1+ka, k') \quad \text{giving} \quad \text{cn}(1+k, k) = \frac{(1+k)(1+ka, k')}{1+ka(1+k)};$$

$$\frac{1}{\text{cn}(2+k, k)} = \frac{1}{1+k} + a(1-k) \frac{1-x}{1+x}, \quad \text{cn}(2+k, k) = \frac{1+k}{1+k} \frac{1+ka}{1+ka(1+k)};$$

$$\frac{\text{dn}(1+k, k)}{\text{cn}(1+k, k)} = \frac{1}{1+k(1+ka)} \frac{(1-x)a}{a+1}, \quad \text{dn}(2+k, k) = \frac{1-ka}{1+ka};$$

where, as before, a, c, d denote $\text{sn } v, \text{cn } v, \text{dn } v$, k modulus k , in fact, the formulæ of the second line of the table, *Elliptic Functions*, p. 63.

Fixed Curve the Cubic $y^2 = x(1-x)(1-k^2x)$. Art. Nos. 138 to 142.

138. It was shown, No. 63, how for the affected theorem, when the fixed curve is a cubic, the form of differential is the multiplied by

$$\overline{012} = \int_a^x [\chi d\omega]_3, (d\omega)_1, (d\omega)_2, \dots],$$

the last term being the properly determined constant K , attached to the variable term $\overline{012}$, in order to obtain a standard form of integral. The object of the present articles is to show what these formulæ become for elliptic integrals of the second and the third kinds respectively.

139. We have, in general,

$$\overline{012}, \overline{012} = \begin{vmatrix} (1+k)x_1y_1, & -(y_1+y_2)y_1 \\ \xi(y_1) + (\xi+1)x_2, & (c_0, c_1, x_1) \\ y_1(y_1)x_2, & \end{vmatrix}, \quad \begin{vmatrix} x, & y, & 1 \\ x_1, & y_1, & 1 \\ x_2, & y_2, & 1 \end{vmatrix}.$$

[book p. 192] Taking here $2, = \emptyset$, for the point, coordinates $x_1, y_1 = 0, 0$, we have

$$\overline{012} = -\frac{x_1y_2}{y_2(x_2-x_1)(y_2-y_1)} = -\overline{012},$$

and, retaining 1 to denote the point coordinates (x_1, y_1) belonging to the argument u_1 , and $1'$ for the point belonging to the argument $u_1 + \frac{1}{2}iK'$; the coordinates of $1'$ are $\frac{1}{k^2x_1}, \frac{1}{k^2x_1^3}y_1$; and the formulæ become

$$\overline{1'0} = -k^2x_1(x - \frac{1}{k^2}x_1(y-0)y_1), \quad = 1+k^2,$$

the numerator herein multiplied by ν , is

$$= -k^2x_1(x - \frac{1}{k^2}x_1(y-0)y_1) = (-\Omega 1)y,$$

which, substituting for y, x, z as their values in terms of u, v, w , becomes

$$= \frac{0 + \nu^2(\nu - z_1z_2 - z_2z_1)}{(\nu\nu_1 + \nu_2\nu_3)(\nu_2 - \nu_1)} = -k^2x_1;$$

Operating on each side with $\frac{a}{dx_2}, = 0$, we obtain

$$\partial_2\overline{012} = -\partial_2,$$

$\partial_3 \overline{012} = w^2 - 12y + L^2(2y - z) - z_1$. The differentiation being, in fact, that which covers in establishing the fundamental property of the elliptic integral of the second kind. Hence we have

$$Z(u+a, k) = (1 - \frac{R}{8})u + L)(n du,$$

viz. we have

$$ZU = ZU_1 - L^2a - L^2z = L(u - u_1) - LK(u - u_1) = \Omega(\nu - x_1);$$

and thence

$$\partial_x ZU = \partial_x ZU_1 - L = LK^2aU(u - u_1)k(u - u_1) = \Omega'.$$

Observe that, L referring to u , all the variable's might be written U' .

140. Continuing the discussion, the expression for the right-hand side is $= \omega + (a-2)\omega'$, and this divided by u^2 , the equation $\frac{1}{2} = 0$. We have here

$$\partial_x \omega - \omega + \omega' = \overline{0(u-u_1)} = (u_1 - u_2)(u_2 - u_3)(u_3 - u_1);$$

viz. this is, $\overline{u_1} = -\overline{u_2}$, in $\overline{u_1}$, in $\overline{u_2}$. From this, and adding, the expression in question becomes

$$\partial_x w(z_1 - z_2) \frac{x_1}{a_1 a_2 a_3} = -\partial_x w(z_1 - z_2) \frac{x_1}{a_1 a_2 a_3},$$

the differentiation being, in fact, that which covers in establishing the fundamental property of the elliptic integral of the second kind.

[book p. 193] **140.** To introduce into the formula 1 instead of $1'$, we have only to write $u_2, -K'$ instead of u_1 , putting also for shortness c, d, f, g, h for the functions of u and a , respectively, we then obtain

$$[01'] = -\frac{x}{y} a_1 (c_1 + x a_1),$$

where observe that, interchanging u and a_1 , we have

$$[110] = \frac{a_1 d_1}{c_1 (c + x_1 c_1)},$$

that is, $[10] = -[01]$, as it should be. The formula may be written

$$[01'] = -[10'] = -\frac{a a_1 d}{x c_1 (c + a a_1)},$$

and

$$\partial_y [01'] = -\frac{1}{(c + a a_1) a a_1},$$

whence

$$\partial_x [10'] = -\frac{1}{a^2 (x - x_1) a c (c + a a_1)},$$

we have, moreover,

$$[012] = [12'] - \frac{x}{y_1} [\overline{036}] - \frac{x_1}{y_1} [306],$$

and

$$\partial_x [012] = \frac{x_1 y_1}{a_1 a_1 (x - x_1)} - \frac{1}{x_1} \frac{1}{x c_1 (c + a a_1)},$$

which last equation gives $(\partial_x + \partial_x + \partial_x) [012] = 0$, as it should be.

141. Supposing that the differential $d\Pi_a$ is defined by the equation

$$d\Pi_a = du_1 [012] + du_2 [01'] + du_3 [026] - [126],$$

we have

$$\int d\Pi_a = du_1 [012] + du_2 [01'2'] + du_3 [026] - [126];$$

and thence

$$\begin{aligned} \partial_x \int d\Pi_a &= \partial_x [012] + du_2 [126] - \partial_y [126]; \\ &= [126] - [126], \\ &= \frac{1}{(c + a a_1)} + \frac{(1 + k^2 a_1 - 2 a_1) a_1}{(1 + k a_1)(1 + k a_2)}; \\ &= \frac{1}{(c + a a_1)} - \frac{1}{(1 + k^2 a_1)} (1 + k a_2)(1 + k a_3). \end{aligned}$$

C. XII.

25

[book p. 194] or establishing between the constants u, a , the relation $\text{sn}^2(u_a - u_1) = \frac{K}{K - R}$, this becomes

$$= 1 + \frac{R}{R - k} k^2 \text{sn}^2(u_a - u_1) a k^2 K^2,$$

which is

$$= \partial_x \log \frac{\Theta(u_1 + a + iK') \Theta(u - u_1 - iK')}{\Theta(u_1 + iK') \Theta(u - iK')},$$

where Θ is Jacobi's theta-function, see my *Elliptic Functions*, p. 144. The expression is, in fact, $= \frac{1}{2} k \log \Pi(u - u_1, a - u_1)$, where the integral $\partial = \int_0^a \log \Theta$.

But we have

$$\log \Theta(u) = \log \frac{\Theta(0 + iK') \Theta(u - iK')}{\Theta(0 + iK') \Theta(0 - iK')} (1 + k(\dots)) + \int_0^u n du;$$

that is, $\partial = \int_0^a \log \Theta$, and consequently we have $\partial_x [\int_0^a \log \Theta] = \Omega$,

$$= \frac{1}{\sqrt{(\nu_1 - \nu_3)(1 - k^2)}} \log \frac{(1 - k\nu_1)\sqrt{2K}}{(1 - k\nu_1)\sqrt{R}} \frac{LK}{(LK)} - 2k(1 - k^2)(1 - k^2)a(1 - k^2a(1 - k^2)),$$

that is, $\partial = \frac{K}{B - K} \sqrt{1 - k} a$, and consequently we have $\partial_x \sqrt{(1 - k\nu_1^2)(1 - k\nu_2^2)} a$,

$$\begin{aligned} &= -\frac{R}{R - k} \partial_x \sqrt{1 - k^2} a^2 - 2k(1 + k^2)(1 - k^2)a^2 + 2ka^2, \\ &= -\frac{R}{R - k} \partial_x \sqrt{(1 - k^2)(1 - k^2)} a \partial_x \sqrt{1 - k^2} a b a, \end{aligned}$$

The difference of the two functions on the right-hand side is $= -\partial_x (u - a) (1 - k \operatorname{sn}^2(u - a))$, which is $= -\operatorname{sn}^2 u a + \operatorname{sn}^2 a$; and the integral being $= 0$ on the lower limit, we readily see that the expression on the right-hand side is, in fact, the function which is required. The transformation also is given in summary by the form which I had obtained. I consider with this remark.

Fixed Curve the Quartic $y^2 = x(1 - x^2)(1 - k^2)$. Art. Nos. 145 to 145.

143. This is a curve having a tacnode at infinity on the line $w = 0$; as may be seen by writing the equation in the homogeneous form $y^2 w^2 = x(1 - x^2)(1 - k^2 x^2)$, and putting for shortness $X = w(1 - x^2)(1 - k^2 x^2)$, $X = w(1 - x^2)(1 - k^2 x^2)$, when X, w small as compared to a , the equation is reduced to $y^2 w^2 = 1$, viz. this is a $\left(\frac{yw}{1}\right)$ to which the sign $\pm$ is to be understood accordingly. I consider with

[book p. 195] reference to this curve only the affected theorem, in the particular form in which it must readily connects itself with the ordinary theory of the integral of the third kind.

144. I consider the differential $\frac{(x, y, z)_p d\omega}{012}$ in the particular case where the line 012 is a line parallel to the axis of y : taking its equation to be $x - x_1 = 0$, and putting for shortness $X = \sqrt{1 - x^2} \sqrt{1 - k^2 x^2}$, $X_1 = \sqrt{1 - x_1^2} \sqrt{1 - k^2 x_1^2}$, the parametric points are taken to be (x_1, X_1) , $(x_1, -X_1)$, and the residues are the intersections with the curve $f - X_1 = 0$ of the line $f = 0$, viz. this is the line passing through the two points of contact of the parallel tangents: the points (x, y) and the points (x_1, X_1) , $(x_1, -X_1)$ are on a conic Ω , containing the affected theorem, and we have for the form $a(x - x_1)$, containing the arbitrary constant a . We have moreover, in usual notation, for the form a containing the arbitrary constant a .

If the limits are taken to be points on the curve $a^2 + y^2 + ax + by + c = 0$, what we want to do is to express the summation of the integrals in the form

$$\int \frac{\sqrt{X}}{x - x_1} \frac{x - x_1}{X} dx + a \int \frac{\sqrt{X}}{x - x_1} \frac{dx}{X} - a \int \frac{\sqrt{X}}{x - x_1} \frac{dx - dx_1}{x_1 X};$$

where $X_1 = \sqrt{1 - x_1^2} \sqrt{1 - k^2 x_1^2}$, etc. Reference to the right curve as above $a^2 + y^2 = ax + b$, when $a^2 + y^2 - 2ax - a^2 = 0$; the right-hand side, by adding the like quantities for the other points a, β, γ .

We have

$$-\frac{1}{a^2} (a^2 - X^2 - a)^2 = \frac{a^2 - aB}{a^2 + a^2 X^2},$$

and the integral thus becomes

$$= -2 \int \frac{\sqrt{X} a d\omega}{a^2 - aB} \frac{a^2 - aB}{a^2 - aB^2 X^2} = a \int \frac{X^2}{a^2 - aB^2 X^2} \frac{dx}{\sqrt{X}}.$$

25-2

[book p. 196] Taking here $a = 0, -c$, so $a = (X^2)$, $a = \frac{a}{\sin \theta}$, we have for such a $\Pi(a, k)$, and the result is

$$\int \frac{\sqrt{X}}{x - x_1} \frac{dx - dx_1}{X} = \int \frac{X dx}{x - x_1 + b a a X^2} - \frac{X a a X}{x - x_1} \sin \theta du,$$

where on the right-hand side the first term is $-\int(x, a, c)$ in Jacobi's form of the integral of the third kind, viz. *Elliptic Functions*, p. 146.

And thence

$$\partial_x \Pi(u, a) = \frac{2(\nu - 1)(z a) a + X(z a + 4u^2 - X a^2)(\nu + 2)(\nu + 2X)}{=} \left(1 - \frac{R}{R}\right),$$

or

$$\partial_z \Pi(u, a) = \frac{R^2(s^2 + r^2) a^2 - 2k^2(1 + k^2) k^2}{(1 - k^2 \nu^2 a) a^2} = \left(1 - \frac{R}{R}\right) + 1 x a^2;$$

or, this being represented in regard to u , (a , we have

$$\partial_z \Pi(u, a) = \omega a + L(0 + a - 2 a du);$$

and thence by integration, and a proper determination of the constants, $\Pi(z, a) = \Pi(z, a)$.

Chapter VI. The Nodal Quartic.

Nodal Quartic; the General and Polynomial Forms. Art. Nos. 146 to 148.

146. For a cubic, or when an x is, the nodal quartic is the next case as regards points on the curve, and corresponding theories with functions of a single argument (elliptic function): but for a node it is the next case as regards the number of points on the curve $\delta = 4, p = 8$; but it is, as in the case of the cubic, the next case as regards the number of points or, with a marked change in the form of the result.

[book p. 197] The most simple curve of deficiency 2 is the nodal quartic, $n = 4, p = 2$. Using homogeneous coordinates, the general form is $Ax^4 + 8Bx^3 + (\dots) = 0$. Using

$$A = (b, j, b)(y, z)^2,$$

$$B = (b, j, b)(y, z)^3,$$

$$C = (p, q, r, s)(y, z)^3,$$

and where we write also

$$D = AC^2 - 2(p, q, r, s)(y, z)^4 + 4(c, j, k,)(y, z)^4 - 4(c, j, k,)(y, z)^4.$$

Clearly the equation of the two tangents at the node is $A = 0$; and the coordinates of the six tangents from the node are $a - kx + by + cz = 0$; $a - kx + by + cz = 0$, the equation of the curve corresponds to the form: that is the case is unaltered if for β, γ, δ we substitute $\alpha + \delta, \beta + b, b + b, b + b$, and we have thus, for the conic from the tangents to the curve. $a' + Y' + Z' = 0$, where $\delta_1, \delta_1, \delta_2$, and δ_2 for the conic, when the cubic, the six tangents are entered by; or, in like manner, we have, $a' + Y' + Z' = 0, a + b + c = 0, b + e + d = 0$, the lines $\delta_1, \delta_2, \delta_3, \delta_2$, are corresponding points.

147. An important special case occurs when $B = 0$: we here have

$$\begin{aligned} A &= (a, h, i, h)(y, z)^2, \\ B &= 0, \\ C &= (p, q, r, s)(y, z)^3 = (a, b, c, d)(y, z)^3, \end{aligned}$$

or, omitting the factors i and j ,

$$(a + j)(b + j)(c + j) = (a + j)(b + j)(c + j)(a + j) = 0.$$

The origin is here a Sencocode; the tangents are $a - z + y = 0$, and there are six tangents to the cubic on the curve.

148. The general nodal depends on 11 constants, but by writing for $uby, uby + cy + dz, ukx + by + cy + ay$, we can reduce the number is like manner to $1 - 1 = 0$. Similarly the Reformatted form depends on 7 constants, but we can reduce the number is like manner to $1 + 1 - 1 = 0$, so that the final form might be taken to be

$$xy = (a - b)(a + b)(a - b)(b - a),$$

this line will be a tangent of

$$(yM + 1)(1\sigma M^2 + 1) = 0;$$

[book p. 198] but it is more convenient to retain the original form

$$x^2(ax + by)^2 = (a - x)(b - y)(c - x)(d - y),$$

bearing in mind that this is reducible to the form just referred to, and thus depends visually upon only 5 constants.

It is a general property that a curve of deficiency p greater than 1 can be by a rational transformation reduced to a curve of the corresponding deficiency $4p - 3$ (this number = 1 for $p = 2$), so far as it can be reduced; and the parametric in particular, $p = 2$, the nodal quartic is to be reduced to a curve $a = 0, \beta = 0$ corresponding properly, viz. the sextic. This will be done in the present chapter, but the form here adopted is not the standard form generally adopted, but the standard form first adopted by Weierstrass as a canonical form.

Reduction to the Plefonoidal Form. Art. Nos. 149 to 152.

149. Consider the general nodal quartic $Ax^4 + 2Bx^3 + (\dots) = 0$. The equation of the tangents from the node $(x, y, z = 1, 0, 0)$ to the curve is the discriminant in regard to ξ of $Ax^4 + 2Bx^3 + (\dots) = 0$. We may write

$$A_x = (a, h, j, h)(y, z)^2, C_x = (p, q, r, s)(y, z)^3$$

and the transformed equation is

$$Ax^4 + 2Bx^3 + (\dots) = A_x x^2 + 2B_x x + C_x = 0, \text{ \&c.}$$

say this equation is

$$Ax^2 + 2Bx + C = 0,$$

where A, B , &c. have the values of A_x, B_x , &c., written in x, y, z , in place of x', y', z' ; and the transformed equation $\Lambda x^2 - 2BxC = 0$ is the equation of the six tangents.

150. From the values of A, B, C we obtain after a little reduction the following formulæ:—

$$A' = (a, h, j, h)(y, z)^2,$$

$$B = (l, m, n, l, j, k)(y, z)^3,$$

$$C = (a, b, c, d, e, f, g)(y, z)^4.$$

and the transformed equation is

$$Ax^2 + 2Bx + C = 0,$$

where

$$\mathbf{A} = A',$$

$$\mathbf{B} = A^2C - (B^2),$$

$$\mathbf{C} = AC^2 - 2(B)^2 + (B^2)C + 2R,$$

[book p. 199] and thence

$$\mathbf{B}^2 - 4\mathbf{A}\mathbf{C} = A^2[A(AC^2 - 2(B)C^2 + (B^2)Cb)] - AC(A(Cbcebd));$$

We have here $B, = A(p, q, r, s)(y, z)^3$, a cubic function $(y, z)^3$ containing the factor $A_x, = (a, h, j, h)(y, z)^2$, and similarly for a, a, a, b . Calling the other factors B, V , we have $B = AB', V = AV'$, respectively; where b, b, c are merely the roots of the equation $AV = 0$.

We have what precedes, by the transformation $x = y + \beta_1, a = \beta, y = \beta$, passed from the form $Ax^4 + 2Bx^3 + Cx = 0$ to the form

$$\mathbf{A} = A, \mathbf{B} = V, \mathbf{C} = -V'(B, \mathbf{\Pi} - C),$$

where

$$\mathbf{\Pi} = (\xi, \eta, k)(y, z),$$

and thus the equation is $Ax^2 - 2Bx + C = 0$.

We have by what precedes, by the transformation, the canonical form $a = 0, y = 0$, viz. $\mathbf{\Pi}$ if we $\mathbf{\Pi}$ have the common factor $A(\mathbf{\Pi} - e^\eta \mathbf{\Pi} - 0)$.

152. Assume now

$$\Sigma B = AB', V = AV'; \mathbf{\Pi} = AB',$$

and therefore also

$$A = (t, u, b\xi y b)(y, z);$$

$$\mathbf{\Pi} = (\xi, \eta, k)(y, z)(xya);$$

$$\mathbf{\Pi} = AC^2 - 2(B)C^2 + (B^2)C;$$

viz. this equation is, the transformation, that of a curve of the order 5 ($p = 2, w, C$); and the equation thus becomes

$$Ax^2 - 2Bx + C = (xy + \mathbf{\Pi}\mathbf{\Pi}); \quad \text{the equation must become a tangent at}$$

and there is to the same equation $a = (\xi, \eta)(y, z)$,

$$xy = (\xi + \eta)(y, z)(xya);$$

where

$$\Xi = (c, f, g)(y, z);$$

$$\Pi = (\xi, \eta, k)(y, z); \quad \&c.;$$

151. Assume that the curve transforms by the substitution $x = y + Z\eta$. We have the change of Π, Ec, z, f, y , in A, C, b, y, c ; and thus

$$\Pi = Aa^2, \quad \Pi = Aa^2, \quad Z = -Cy^2 + AZ^2 + A\eta\xi(LVa);$$

which putting therein $(\xi, \eta, k)(y, z)$, the equation must become

$$Y = (a, b, c, d, e, f, g)(y, z);$$

and therein removed by

$$\Xi, Y = Z + Mp; \quad Y = Z + K - Mp - q;$$

and therefore would be in the in $Y Y$.

[book p. 200] then in the new coordinates (X, Y, Z) we have the equation

$$(i, j, k)(Y, Z)YZ + [(X - aY - bZ)^2 - X(p, q, r, s)(Y, Z)] - kZ$$

$$+ (1, b, c, d, k)(X, Y)(Y - 2bZ - kY) + (X X(Y - bZ)(X - aY - bZ)(X - aY - bZ)(X - aY - bZ)$$

$$+ 4(X - aY - bZ)(X - aY - bZ)(X - aY - bZ)(X - aY - bZ)(X - aY - bZ)(X - bZ - kY) = 0,$$

that is,

$$(t, j, k)(Y, Z)YZ + [(X - bY - bZ)^2 - X(p, q, r, s)(Y, Z)]$$

$$+ 4(X - aY - bZ)(X - aY - bZ)X(X - bY - bZ) = 0;$$

$$+ 1(X - bY - bZ)(X - aY - bZ)(X - bY - bZ)(X - bY - bZ)(X - bZ - kY) = 0;$$

when this equation $(X, Y, Z) = 0, Y(X - bY - bZ) = 0$, the equation transformations must become

$$(X - aY - bZ)(X - aY - bZ)(X - aY - bZ)(X - aY - bZ)$$

$$+ 4(X - aY - bZ)(X - aY - bZ)(X - aY - bZ)(X - aY - bZ)(X - aY - bZ) = 0,$$

that is,

$$(p, q, r, s)(Y, Z)^4 - (Y - 2bZ - kY)^2(X - bY - bZ - bZ - kY)^2$$

$$- (X - aY - bZ)^2(X - aY - bZ)^2(X - aY - bZ)^2(X - aY - bZ)^2(X - aY - bZ - kY)^2 = 0,$$

or rearrangely

$$X, Y, Z = X - aY - bZ, \quad Lz = Mb(bY - bZ) - kY,$$

transformed into the differential form as above.

It is similarly the case that the differential first appears, although there is a great many more steps to be made in establishing the truth of the previous, and the case where the case the parametric points are a curve of deficiency p , p being p .

Application of Abel's Theorem.

152. Taking the fixed curve to be $f, = (Ax^2 + 2Bx + C) = 0$, we have

$$\frac{df}{dx} = Ax + B;$$

if for shortness we write

$$(x, y)^2 = P, = P, Bx + Cy^2, y = (Ay - Cz)(z - y)P;$$

[book p. 201] *Application of Abel's Theorem.* Art. Nos. 153 to 157.

152. Taking the fixed curve to be $f, = (Ax^2 + 2Bx + C) = 0$, we have

$$\frac{df}{dx} = Ax + B,$$

if for shortness we write

$$(x, y)^2 = P, = Ax^2 + 2Bxy + Cy^2, P_x = (Ay - Cz)(y - z)(z - y) = (Ax + B)(\dots),$$

and we thence have

$$d\omega = \frac{x dy - y dx}{\sqrt{P}}.$$

The minor curve $(x, y, z)^{n-3} = 0$ is an arbitrary line passing through the node, that is, the point $x = 0, y = 0$; and there passes through this line one of two distinct relations $2x dz - dx = M(x dz - dz), = 0$, viz. this is an ordinary line through the points 1 and 2 of the curve.

The variable curve is taken to be an arbitrary curve passing through the intersections of the conic Ξ , with the conic $\mathbf{X}$ passing through the four points belonging to these values $2c, b, a, c$. Specifically the line is arbitrary, and the line being given, the equation determines the intersections of the line with $\mathbf{X}$ passing through these intersections of the line with $\mathbf{X}$ which are the points 1, 2, 3, 4, viz. we have

$$\frac{df}{dy} = Ay + B, \quad Lz = Mp, \quad LZ = K - Mp;$$

to complete the verification, observe that we have $Y = 8TX^2 - (1 + 8T^4)X$, whence

$$-T = (1 + 8T^4)(-YX + 8(PY - X^4) - 16YT^4 - 8X^4 = 16T^4X^4.$$

153. Observe that the intersections of the cubic with the fixed curve are given by the equation $\Omega + \mathbb{P}\mathbf{X}\mathbf{\Pi} = 0$, viz. $\Omega + \mathbb{P}\mathbf{X}\mathbf{\Pi} a, b, c, d, e, f, g)(x, y, z)^4$, an equation which determines a ; and the ratio $x : y$ is then determined rationally by the equation $Ax - By = 0$. Or what is the same thing, we may write

$$A = a^2 + 8aBy^2 = 6aBy^2, \quad B = -a^2y^2 + (1 + 16y^2 + 8y^2)8(8a + 8BX) + 8(15)yXX,$$

in like manner, the four equations are given when the points 1 and 2 are determined. Conversely if (y, y) and (z, z) are corresponding points the conic $\Omega + 0X^2$ pass through the points 1, 2, 3, 4.

154. Observe that the intersections of the cubic with the fixed curve are given by the equation $\Omega + \mathbb{P}\mathbf{X}\mathbf{\Pi} = 0$; the line $\mathbf{\Pi} = 0$, hence the conic passes through the points 5, 6, 7, 8, where 5, 6, 7, 8 are the intersections of the lines $\xi = 0$ with the conic. The variable line $\mathbf{T}P + \mathbf{X}P = 0$

corresponds to the cubic transformed integral; the deficiency is here equal to 2, integral relations, which agree with the conditions of a curve of deficiency p , p being 2.

[book p. 202] **155.** If we now assume

$$\begin{aligned} d\omega &= x d\omega_1 d\omega_2, & d\omega &= x d\omega_2, & d\omega_2, & d\omega &= x d\omega_2, \\ d\omega &= x d\omega_2 + y d\omega_1, & d\omega &= x d\omega_2 + y d\omega_2, & d\omega &= x d\omega_2 + y d\omega_2, \end{aligned}$$

or say

$$\begin{aligned} u, v &= \int_0^1 (x, y) \left(\frac{x dy - y dx}{\sqrt{P}} + \int_0^2 (x, y) \left(\frac{x dy - y dx}{\sqrt{P}} \right), \\ u', v' &= \int_0^1 (x, y) \left(\frac{x dy - y dx}{\sqrt{P}} + \int_0^2 (x, y) \left(\frac{x dy - y dx}{\sqrt{P}} \right), \end{aligned}$$

where u, v, u', v' are each of them, of course, regarded as functions of the four points 1, 2, 1', 2', then u, v, u', v' are functions which are connected together by the proper relations, and which present themselves either as singly, or simultaneously by means of the formula for σ functions.

In a basically completely determined way, the points 1, 2, 1, 2' are not perfectly determined, but are determined by elliptic integrals: but we have these functions, and the systems of points 1, 2, 1', 2' taken in connection with the various functions (u, v, u', v') in a system as 1, 2, 1', 2', etc. are points which it is the difference between the formula on the integrals. We have, in fact, that

$$\begin{aligned} \sqrt{u(u_1 - u_2) - 2(u_1 + u_2)u'} &= u(u_1 - u_3 + u_2)u'(\dots), \\ B(u_1 - u_2), &= u(u_3 - u_2)A(u_1 - u_3)(\dots), \\ A(u_1 - u_2), &= u(u_3 - u_2)A(u_1 - u_3)(\dots), \end{aligned}$$

and so on for the four other systems of equations $C(u, v)$, $D(u, v)$ respectively, formed in the case of an integral function of u, v of orders 1, 2, respectively (and not in the case of an arbitrary function ρ, ρ' of these integrals).

156. The differential relations thus become

$$du du' + dv dv' = 0, \quad x' du - y' dv = 0, \quad x du - y dv;$$

where T, J are constants which are determinable as definite integrals by the consideration that du, dv form a complete system. The equations $u_1 + u_2 = u, v_1 + v_2 = v$, then the points u, v are points where we have the consideration that the conic passes through the points 1, 2, 1', 2' on the cubic; whence considering the conic as a function of the conic f may be defined as a continuous function of these arguments: there is then a marked change in the integral to the number of constants in a curve of deficiency p .

156. The differential relations thus become

$$du du' + dv dv' = 0, \quad x du + y dv = 0, \quad x' du + y' dv = 0,$$

where J, J are constants which are determinable as definite integrals by the consideration that the conic f may be defined as a continuous function of these arguments: there is then a marked change in the integral to the number of constants in a curve of deficiency p .

[book p. 203] **157.** We may, in the first instance, disregarding altogether the consideration of the arguments u, v , attend only to the algebraic functions such as $\sqrt{X}, \sqrt{Y}, \sqrt{Z}$, of the coordinates of the pairs of points 1, 2, 1', 2', 3, 4, and so on as regards $\mathbf{X}$ as a function of the coordinates of the points only.

Origin of the Double-Letter Functions. Art. Nos. 158 and 159.

158. The cubic curve $Ax + B = 0$ may be taken to be a curve through two of the points of contact, say the points α, β : we have a Ω a function $(x, y)^2$; or throwing out the factor $(z - \beta y)(z - \beta y)y$, by the identity

$$(\alpha - x)(\beta - y) = -y(\alpha - y)(\beta - x) + (a, b, c, d, e, f, g, h)(x, y, z)^2,$$

and hence, introducing $\sqrt{a}, \sqrt{b}$ as quasi-coordinates of the curve, the original a becomes equal to the identity

$$(\sqrt{a} - x)(\sqrt{a} - y) + (a, b, c, d, e, f, g, h)(x, y, z)^2 = 0,$$

and hence, introducing a notation which it is convenient to use, we set

$$\begin{aligned} & \sqrt{a} \sqrt{b} - x \sqrt{a} y \sqrt{b} a (b - x \sqrt{a})(\sqrt{b} \sqrt{c})(c - b \sqrt{c})(b - x \sqrt{a}) a (b - x \sqrt{a})(\sqrt{c} b) \\ & - Y (b - x \sqrt{a})(b - \sqrt{a})(a - x \sqrt{a}) a y (a - x \sqrt{a})(b - x \sqrt{a}) a b (a - x \sqrt{a}) a b a b; \end{aligned}$$

159. Putting each of the y 's equal 1, we have the identity

$$(\alpha - x_1)(\alpha - x_2) - 2(1 + x_1)(1 + x_2)(1 - x_1)(1 - x_2)(1 - x_3)(1 - x_4) = 0,$$

and hence, putting in for shortness

$$a = \sqrt{a(1 + x_1)(1 + x_2)}, \quad x = \sqrt{(1 + x_1)(1 + x_2)x_1x_2}, \quad x' = \sqrt{(1 - x_1)(1 - x_2)x_1x_2};$$

and in like manner, $\beta = \sqrt{b(1 + x_1)(1 + x_2)b'b'a(bb' - c)(cb' - c)b}$, where the symbols $ab a, abb', a a', bcab', bbbca a'$ are determinate.

[book p. 204] We have thus

$$\frac{(a - x_1)(a - x_2)}{(a - x_1)(a - x_2)} = \frac{a^2}{a\sqrt{a}a\sqrt{a}}, \quad = \frac{ax_1ax_2}{ax_1\sqrt{a}ax_2\sqrt{a}ax_1ax_2} aabacd,$$

which last equation, taking the square root, becomes

$$\frac{a - x_1}{\sqrt{a}ax_1} = -\frac{ax_2ax_3}{\sqrt{a}ax_3ax_4},$$

or, substituting in the first equation, we have

$$\frac{x - x_1}{x - x_1} = \frac{ax_1ay_1ax_2ay_2a}{ax_1ax_2aa^2\sqrt{ay_1ay_2}ax_1\sqrt{ax_2}},$$

which clearly divides out being in each of these cases respectively.

159. Considering the dual DE as an abbreviation for the double letter ABC, DEF , the expressed dual being always accompanied by the letter E where we have left in the consideration of the double-letter functions

$$\Lambda_{ab} = \frac{1}{ax_1\sqrt{a}ax_2\sqrt{a}ax_1\sqrt{ax_2}} a(b\sqrt{ax_1}\sqrt{ax_2}\sqrt{a}a) \cdots,$$

in connexion with the already mentioned single-letter functions $\Lambda_a = \sqrt{ax_1ax_2}$, etc. viz. in this notation the equation just proved is

$$\Lambda_{ab} = \frac{\sqrt{(a-1)(1+a-c)}\sqrt{a-d}}{\sqrt{(a-1)(1+a-c)}\sqrt{a-d}} \frac{\Lambda_a\Lambda_a}{\Lambda_a};$$

and it thus appears that, by reason of the constancy of the symmetry of the first three letters and of the symmetry of the last three letters in the function Λ_{ab} , Λ_{ab} , the function Λ_{ab} , a , b is interchangeable by the suffices a and b , c , equal to that of the single-letter functions.

160. Considering the dual DE as an abbreviation for the double letter ABC , DEF , the expressed dual being always accompanied by the letter E where we have left in the consideration of the double-letter functions $\Lambda_{ab} = \sqrt{ax_1}\sqrt{ax_2}$, &c.

In connexion with the already mentioned single-letter functions $\Lambda_a = \sqrt{ax_1 ax_2}$, etc. we have a system of formulæ as $\sqrt{ax_1 ax_2}$, etc. and it is from these formulæ were considered by Aronhold in his (Göttingen) Memoir of 1865 (whence the term “double-letter functions”). It is in like manner be shown that the function $\Lambda_a = \sqrt{ax_1 ax_2}$, etc. is the quotient of a major function $(x, y, z)^{n-2}$ by the linear function 012.

Chapter VII. The Addition Theory.

The Addition Theory. Art. Nos. 160 to 163.

160. We have the six single-letter symbols A, B, C, D, E, F , viz. $\Lambda_a = \sqrt{(a-x_1)(a-x_2)}$, etc. and the ten double-letter symbols $AB, AC, AD, AE, AF, BC, BD, BE, CD, CE, DE$, viz. $\Lambda_{ab} = \frac{1}{a-b} [\sqrt{(a-x_1)(b-x_2)} - \sqrt{(a-x_2)(b-x_1)}]$, etc. from 10 functions being connected by algebraical relations which admit immediately deducible from the considerations of the various forms of $\Lambda_{ab} \Lambda_{ab}$, etc. The relations between the variables 1 and 2 being so connected that the systems form, in regard to one of these variables, say x_1 , an integral relation, which agrees with the symbol a , of the function above considered.

The relation between the variables x_1, x_2 is that they correspond to a line meeting the curve in two points which we call 1, 2, viz. taking the equation in the form

$$(ax^2 + 2bx + c)y^2 - (a-x)(b-x)(c-x)(d-x) = 0,$$

where it is a character expression of the function $(x, y, z)^{n-2} = 0$, of the line is to be taken to be

$$ay^2 + bx^2 + 2c'y + d = 0,$$

and we have thus the two relations

$$(ax_1^2 + 2bx_1 + c)y_1^2 = (a-x_1)(b-x_1)(c-x_1)(d-x_1),$$

$$(ax_2^2 + 2bx_2 + c)y_2^2 = (a-x_2)(b-x_2)(c-x_2)(d-x_2),$$

$$ay_1y_2 + bx_1x_2 + c'(x_1 + x_2) + d = 0;$$

which equations serve to determine the ratios $a : \beta : k$ in terms of x_1, x_2, y_1, y_2 , and we have then the two relations

$$ax_1y_2 + bx_2y_1 + 2c'y_2 + d(\dots),$$

$$ay_1 + bx_1 + 2c'y_2 + d(\dots).$$

which determines the symmetric functions of x_1, x_2 .

[book p. 205] *The Addition Theory.* Art. Nos. 160 to 163.

160. We have the six single-letter functions A, B, C, D, E, F , viz. $\Lambda_a = \sqrt{(a-x_1)(a-x_2)}$, &c.; and the ten double-letter symbols $AB, AC, AD, AE, AF, BC, BD, BE, CD, CE, DE$, viz. $\Lambda_{ab} = \frac{1}{a-b} [\sqrt{(a-x_1)(b-x_2)} - \sqrt{(a-x_2)(b-x_1)}]$, &c.

These 10 functions being connected by algebraical relations which admit immediately deducible from the consideration of the various forms of $\Lambda_{ab} \Lambda_{ab}$ etc. The relations between the variables 1 and 2 being so connected that the systems form, in regard to one of these variables, say x_1 , an integral relation, which agrees with the symbol a , of the function above considered.

161. The relation is, in fact, as is given in any of the papers 1, 2 &c. of the double- $\mathbf{T}$ -function, Crelle, t. LXXVIII. (1865), pp. 74–81, [785]. Writing successively $z = a, \beta, b, b, b, a, \beta, b, b', a, \beta, b$, we have

$$\begin{aligned} ax_1^2 + 2bx_1 + c &= y_1 \sqrt{(a-x_1)(b-x_1)(c-x_1)(d-x_1)}, \\ ax_2^2 + 2bx_2 + c &= y_2 \sqrt{(a-x_2)(b-x_2)(c-x_2)(d-x_2)}; \end{aligned}$$

which equations serve to determine the ratios $a : \beta : k$ in terms of x_1, x_2, y_1, y_2 , and we have then the two relations

$$\begin{aligned} ax_1y_2 + bx_2y_1 + 2c'y_2 + d(\dots), \\ ay_1 + bx_1 + 2c'y_2 + d(\dots), \\ ay_1 + bx_2 + 2c'y_2 + d(\dots), \\ ay_1 + bx_1 + 2c'y_2 + d(\dots). \end{aligned}$$

which equations serve to determine the ratios $a : \beta : k$ in terms of x_1, x_2, y_1, y_2 , and we have then the two relations

$$\begin{aligned} ay_1 + bx_1 + 2c'y_2 + d(\dots), \\ ay_1 + bx_2 + 2c'y_2 + d(\dots), \end{aligned}$$

which determines the symmetric functions of x_1, x_2 .